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Stable Angina

Full Review: Jul 2026 By Ranya N. Sweis, MD, MS, Northwestern University Feinberg School of Medicine | Arif Jivan, MD, PhD, Northwestern University Feinberg School of Medicine | Peer reviewed by Jonathan G. Howlett, MD, Cumming School of Medicine, University of Calgary
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Angina pectoris is a clinical symptom of precordial discomfort or pressure due to transient myocardial ischemia without infarction. Stable angina is angina that is typically precipitated by exertion or psychologic stress and relieved by rest or sublingual nitroglycerin. Diagnosis is by symptoms, electrocardiography, and myocardial imaging. Treatment may include antiplatelet medications, nitrates, beta-blockers, calcium channel blockers, angiotensin-converting enzyme inhibitors, statins, and coronary angioplasty or coronary artery bypass graft surgery.

Etiology of Stable Angina

Angina pectoris occurs when:

- Myocardial oxygen demand exceeds the ability of coronary arteries to supply an adequate amount of oxygenated blood

Such imbalance between supply and demand can occur when the arteries are narrowed. Narrowing usually results from:

- Coronary artery atherosclerosis

Narrowing of the coronary arteries can also result from:

- Coronary artery spasm
- Coronary artery embolism (rare)

Acute coronary thrombosis can cause angina if obstruction is partial or transient, but it usually causes acute myocardial infarction (MI).

Because myocardial oxygen demand is determined mainly by heart rate, systolic wall tension, and contractility, narrowing of a coronary artery typically results in angina that occurs during exertion and is relieved by rest.

In addition to exertion, cardiac workload can be increased by disorders such as hypertension, aortic stenosis, aortic regurgitation, or hypertrophic cardiomyopathy. In such cases, angina can occur whether atherosclerosis is present or not. These disorders can also decrease relative myocardial perfusion because increased myocardial mass and stiffness can reduce diastolic flow through the coronary arteries.

A decreased oxygen supply, as in severe anemia or hypoxia, can also precipitate or aggravate angina.

Pathophysiology of Stable Angina

Angina may be:

- Stable
- Unstable

In **stable angina,** the relationship between workload or demand and ischemia is usually relatively predictable.

Unstable angina is clinically worsening angina (eg, angina at rest or with increasing frequency and/or intensity of episodes) and is considered an acute coronary syndrome.

Atherosclerotic arterial narrowing is not entirely fixed; it varies with the normal fluctuations in arterial tone that occur in all people. Thus, more people have angina in the morning, when arterial tone is relatively high (1, 2). Also, abnormal endothelial function may contribute to variations in arterial tone; for example, in endothelium damaged by atheromas, stress of a catecholamine surge causes vasoconstriction rather than the normal vasodilatory response (3).

As the myocardium becomes ischemic, coronary sinus blood pH falls, cellular potassium is lost, lactate accumulates, ECG abnormalities appear, and ventricular function (both systolic and diastolic) deteriorates. Left ventricular (LV) diastolic pressure usually increases during angina, sometimes inducing pulmonary congestion and dyspnea. The full mechanism by which ischemia causes discomfort is unclear but may involve afferent nerve fiber stimulation by hypoxic metabolites (4).

Pathophysiology references

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Symptoms and Signs of Stable Angina

Angina may be a vague, barely troublesome ache or may rapidly become a severe, intense precordial crushing sensation. It is rarely described as "pain." Discomfort is most commonly felt beneath the sternum, although location varies. Discomfort may radiate to the left shoulder and down the inside of the left arm, even to the fingers; straight through to the back; into the throat, jaws, and teeth; and, occasionally, down the inside of the right arm. It may also be felt in the upper abdomen.

Typical angina classically has the following characteristics (1):

- Substernal location
- Exacerbated by exercise or emotional stress
- Relieved by nitroglycerin

"Atypical" symptoms, including bloating, gas, abdominal distress, or burning or tenderness in the back, shoulders, arms, or jaw, may occur in some patients. Patients often ascribe these symptoms to indigestion; belching may even relieve the symptoms. Other patients have dyspnea due to the sharp, reversible increase in LV filling pressure that often accompanies ischemia. Frequently, the patient’s description is imprecise, and whether the problem is angina, dyspnea, or both may be difficult to determine.

Although patients who present with atypical symptoms are generally thought to have a lower likelihood of cardiac ischemia than those with typical symptoms, females with cardiac ischemia do commonly have atypical symptoms such as epigastric symptoms, palpitations, and discomfort in locations other than substernal (1). Furthermore, females with angina are at a higher risk than males for underdiagnosis of cardiac causes. For these reasons, a high index of suspicion for cardiac ischemia should be maintained when females present with atypical symptoms.

Older patients with cardiac ischemia may present with dyspnea, syncope, unexplained falls, or delirium (1).

Because ischemic symptoms require a minute or more to resolve, briefer, fleeting sensations rarely represent angina.

Between and even during attacks of angina, physical findings may be normal. However, during the attack, heart rate may increase modestly, blood pressure (BP) is often elevated, heart sounds become more distant, and the apical impulse is more diffuse. The second heart sound (S2) may become paradoxical because LV ejection is more prolonged during an ischemic attack. A fourth heart sound (S4) is common, and a third heart sound (S3) may develop. A mid or late systolic apical murmur, shrill or blowing—but not especially loud—may occur if ischemia causes localized papillary muscle dysfunction, causing mitral regurgitation.

Stable angina is typically triggered by exertion or strong emotion, usually persists no more than a few minutes, and subsides with rest. Response to exertion is usually predictable, but in some patients, exercise that is tolerated one day may precipitate angina the next because of variations in arterial tone. Symptoms are exaggerated when exertion follows a meal or occurs in cold weather; walking into the wind or first contact with cold air after leaving a warm room may precipitate an attack. Symptom severity is often classified by the degree of exertion resulting in angina (see table Canadian Cardiovascular Classification System for Angina Pectoris).

TABLE

Canadian Cardiovascular Society Classification System for Angina Pectoris

Class	Activities Triggering Chest Pain
1	Strenuous, rapid, or prolonged exertion
	Not usual physical activities (eg, walking, climbing stairs)
2	Walking rapidly
	Walking uphill
	Climbing stairs rapidly
	Walking or climbing stairs after meals
	Cold
	Wind
	Emotional stress
3	Walking, even 1 or 2 blocks at usual pace and on level ground
	Climbing stairs, even 1 flight
4	Any physical activity

Class	Activities Triggering Chest Pain
	Sometimes occurring at rest

Campeau L. The Canadian Cardiovascular Society grading of angina pectoris revisited 30 years later. *Can J Cardiol.* 2002;18(4):371-379; Ferraro R, Latina JM, Alfaddagh A, et al. Evaluation and Management of Patients With Stable Angina: Beyond the Ischemia Paradigm: JACC State-of-the-Art Review. *J Am Coll Cardiol.* 2020;76(19):2252-2266. doi:10.1016/j.jacc.2020.08.078

Attacks may vary from several a day to symptom-free intervals of weeks, months, or years. Attacks may increase in frequency (called crescendo angina), leading to MI or death. Conversely, attacks may gradually decrease or disappear if adequate collateral coronary circulation develops, the ischemic area infarcts, or heart failure or intermittent claudication supervenes and limits activity.

Nocturnal angina occurs with changes in respiration, pulse rate, and BP sufficient to cause myocardial oxygen supply-demand mismatch. Contributing factors may include normal autonomic changes during sleep (particularly increased sympathetic tone and altered coronary blood flow in rapid eye movement [REM] sleep) and sleep disordered breathing in patients with pre-existing coronary stenosis (2). Nocturnal angina may also be a sign of recurrent LV failure, an equivalent of nocturnal dyspnea. The recumbent position increases venous return, stretching the myocardium and increasing wall stress, which increases oxygen demand.

Angina decubitus is angina that occurs spontaneously when a person is lying down but not necessarily at night. It is usually accompanied by a modestly increased heart rate and a sometimes markedly higher BP, which increase oxygen demand. These increases may be the cause of rest angina or the result of ischemia induced by plaque rupture and thrombus formation. If angina is not relieved, unmet myocardial oxygen demand increases further, making MI more likely. Pericarditis is also in the differential diagnosis for chest pain that worsens in the supine position, particularly if sharp.

Unstable angina

Because angina characteristics are usually predictable for a given patient, any changes (ie, angina at rest, new-onset angina, increasing angina, new nocturnal angina, or new angina decubitus) should be considered serious, especially when the angina is severe (ie, Canadian Cardiovascular Society class 3 or 4). Such changes are termed unstable angina and require prompt evaluation and treatment.

Symptoms and signs references

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Diagnosis of Stable Angina

- History and physical examination
- Electrocardiography (ECG)
- Stress testing with ECG or imaging (using echocardiography, radionuclide imaging, positron emission tomography [PET], or MRI)
- CT angiography or CT fractional flow reserve (CT FFR)
- Coronary angiography for significant symptoms, positive stress test, or significant lesions noted on CT FFR

Diagnosis of angina is suspected if chest discomfort is typical, including being precipitated by exertion and relieved by rest, or if symptoms are atypical. Presence in the history of significant risk factors for coronary artery disease (CAD) adds weight to reported symptoms, even if atypical. A high index of suspicion for CAD is maintained when females report atypical (or typical) symptoms.

Patients whose chest discomfort lasts > 20 minutes or occurs during rest or who have syncope or heart failure are evaluated for an acute coronary syndrome.

Chest discomfort may also be caused by gastrointestinal disorders (eg, gastroesophageal reflux, esophageal spasm, indigestion, cholelithiasis), costochondritis, anxiety, panic attacks, hyperventilation, and other cardiac disorders (eg, aortic dissection, pericarditis, mitral valve prolapse, supraventricular tachycardia, atrial fibrillation), even when coronary blood flow is not compromised.

ECG is always performed. More specific tests include stress testing with ECG or with myocardial imaging (eg, echocardiography, radionuclide imaging, PET, MRI), coronary CT angiography, and coronary angiography. Noninvasive tests are considered first.

ECG

If angina is present, especially during exertion, ECG is indicated. Because angina resolves quickly with rest, ECG rarely can be performed during an attack except during stress testing.

ECG is critical in the evaluation of angina to rapidly determine whether patients are experiencing a ST-segment elevation myocardial infarction (STEMI) or a non-STEMI acute coronary syndrome (including unstable angina). ECG can also screen for prior myocardial infarction, arrhythmias, and left ventricular hypertrophy (1).

If performed **during an episode of angina,** ECG is likely to show reversible ischemic changes:

- T wave discordant to the QRS vector
- ST-segment depression (typically)
- ST-segment elevation
- Decreased R-wave height
- Intraventricular or bundle branch conduction disturbances
- Arrhythmia (usually ventricular extrasystoles)

Between episodes of angina, the ECG may be normal, may show evidence of prior infarction, or may show evidence of ischemia.

A normal ECG does not exclude coronary ischemia as a cause for angina (2).

Stress testing

Stress testing is used to:

- Confirm the diagnosis
- Evaluate disease severity
- Determine appropriate exercise levels for the patient
- Help predict prognosis

Exercise stress testing with ECG is performed if a patient has a normal resting ECG and can exercise. Most often it is used in conjunction with additional myocardial imaging. Exercise ECG alone has a sensitivity of approximately 46 to 68% and a specificity of approximately 54 to 77% for obstructive coronary artery disease (1, 3). However, exercise ECG alone can miss severe CAD (even left main or 3-vessel disease). False-positive test results occur, but some of these may represent true ischemic changes caused by microvascular disease (4).

Stress testing with myocardial imaging is usually performed when the resting ECG is abnormal because false-positive ST-segment shifts are common on the stress ECG. Exercise or pharmacologic stress (eg, with dobutamine or dipyridamole infusion) may be used depending on the patient's ability to exercise. Imaging options include stress echocardiography (sensitivity for obstructive CAD of 80 to 89% and specificity of 72 to 89%), myocardial perfusion imaging with single-photon emission CT (SPECT) (sensitivity of 83 to 90% and specificity of 63 to 76%), or PET, and stress MRI (1).

The choice of imaging technique depends on institutional availability and expertise. Exercise ECG remains widely available and inexpensive. Stress imaging can help assess LV function and response to stress; identify areas of ischemia, infarction, and viable tissue; and determine the site and extent of myocardium at risk. Stress echocardiography can also detect ischemia-induced mitral regurgitation.

If the clinical or working diagnosis is unstable angina, early stress testing is contraindicated, except in certain low-risk patients (5, 6).

Noninvasive imaging of the coronary arteries

Coronary CT angiography (CCTA) can accurately identify coronary stenosis and has a number of advantages (1). CCTA is noninvasive, and in patients presenting with stable angina, it can exclude coronary stenosis with high accuracy (sensitivity of 93 to 99% and specificity of 67 to 86%) (1). Noninvasive estimation of the fractional flow reserve (FFR, the ratio of maximal flow through the stenotic area to normal maximal flow) across significant lesions and estimation of lesion-specific ischemia are also possible (2, 7). In addition, CCTA can establish existing stent or bypass graft patency, can show cardiac and coronary venous anatomy, and can assess calcified and noncalcified plaque burden. However, radiation exposure is greater than with other modalities, and CT angiography must be used judiciously in patients with a heart rate of > 65 beats/minute, those with irregular heart beats, those with impaired kidney function, and pregnant patients. Patients must also be able to hold their breath for 15 to 20 seconds, 3 to 4 times during the study.

CCTA is suggested by some as an initial diagnostic test of choice, along with echocardiography, for patients with stable angina, in particular patients younger than 65 years of age with a intermediate or high probability of obstructive CAD (8, 9). It can help categorize patients into those with obstructive and those with nonobstructive disease for further diagnosis and treatment.

Other indications for CCTA include:

- Patients who are asymptomatic but at high risk (10)
- Patients with atypical or typical angina who have inconclusive exercise stress test results, cannot undergo exercise stress testing, or need to undergo major noncardiac surgery
- Patients in whom invasive coronary angiography was unable to locate a major coronary artery or graft

Coronary calcium scoring, obtained via a noncontrast CT scan, can detect the amount of calcium present in coronary artery plaque. The calcium score is roughly proportional to the risk of subsequent coronary events. Coronary calcium scoring is primarily used in screening asymptomatic patients (11), in conjunction with existing risk estimation tools such as PREVENT.

Cardiac magnetic resonance imaging (MRI) is valuable in evaluating many cardiac and great vessel abnormalities. It may be used to evaluate CAD by several techniques, which enable direct visualization of coronary stenosis, assessment of flow in the coronary arteries, evaluation of myocardial perfusion and metabolism, evaluation of wall motion abnormalities during stress, and assessment of infarcted myocardium vs viable myocardium.

Indications for cardiac MRI include evaluation of cardiac structure and function and assessment of myocardial viability. Cardiac MRI, specifically stress perfusion MRI and quantitative myocardial blood flow analysis, may also be indicated for diagnosis and risk assessment in patients with either known CAD or an intermediate to high likelihood of having CAD (10). Stress perfusion MRI has a reported sensitivity of approximately 89% and specificity of approximately 84% for flow-limiting coronary stenosis.

Coronary angiography

Coronary angiography is the gold standard for diagnosing coronary artery disease but is not always necessary to confirm the diagnosis. It is indicated primarily to locate and assess severity of coronary artery lesions when revascularization (percutaneous coronary intervention [PCI] or coronary artery bypass grafting [CABG]) is being considered. Angiography may also be indicated when knowledge of coronary anatomy is necessary to advise about work or lifestyle needs (eg, discontinuing job or sports activities). Although angiographic findings do not directly show hemodynamic significance of coronary lesions, guidewires with pressure or flow sensors can be used to estimate blood flow across stenoses. Blood flow is expressed as fractional flow reserve (FFR, the ratio of maximal flow through the stenotic area to normal maximal flow). If FFR or similar measurements are not available, obstruction is assumed to be physiologically significant when the luminal diameter is reduced > 70%. Angina does not usually develop when the diameter reduction is < 70% unless spasm or thrombosis is present (12).

Intravascular ultrasound (IVUS) provides images of coronary artery structure. An ultrasound probe on the tip of a catheter is inserted in the coronary arteries during angiography. Optical coherence tomography (OCT) is another imaging modality that can be used during coronary angiography that uses near-infrared light to provide high-resolution cross-sectional images of the coronary arteries, which are higher in resolution than with IVUS. Both IVUS and OCT are used to guide stent placement during percutaneous coronary intervention; IVUS is also used to evaluate the extent and significance of some left coronary artery stenoses (6).

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Treatment of Stable Angina

- Acute symptom management with nitroglycerin
- Modification of risk factors
- Lipid management and antiplatelet therapy (aspirin and sometimes clopidogrel, prasugrel, or ticagrelor)
- Beta-blockers
- Sometimes long-acting nitroglycerin and calcium channel blockers for symptom control
- Often angiotensin-converting enzyme (ACE) inhibitors or angiotensin II receptor blockers (ARBs)
- Lipid-lowering agents
- Revascularization if symptoms persist despite medical therapy

This discussion of treatment focuses on patients with stable angina due to chronic obstructive coronary disease. Patients with ischemia and nonobstructed coronary arteries (INOCA), including patients with microvascular angina and vasospastic angina, are discussed elsewhere (1).

Acute symptom management

To relieve symptoms during an acute attack, sublingual **nitroglycerin** is the most effective medication (2, 3). Nitroglycerin is a potent smooth-muscle relaxant and vasodilator. Its main sites of action are in the peripheral vascular tree, especially in the venous or capacitance system, and in coronary blood vessels. Even severely atherosclerotic vessels may dilate in areas without atheroma. Nitroglycerin lowers systolic BP and dilates systemic veins, thus reducing myocardial wall tension, a major determinant of myocardial oxygen need. Sublingual nitroglycerin is given for an acute attack or for prevention before exertion. Dramatic relief usually occurs within 1.5 to 3 minutes, is complete by about 5 minutes, and lasts up to 30 minutes. The dose may be repeated every 4 to 5 minutes up to 3 times if relief is incomplete. Patients should always carry nitroglycerin tablets or aerosol spray to use promptly at the onset of an angina attack. Patients should store tablets in a tightly sealed, light-resistant glass container, so that potency is not lost. Because the medication deteriorates quickly, small amounts should be obtained frequently.

Risk modification

Reversible risk factors are modified as much as possible. People who smoke tobacco should stop smoking; smoking cessation provides important short- and long-term benefits, including a 36% reduction in all-cause mortality after at least 2 years (3, 4). Weight loss alone often reduces the severity of angina. Guideline-based nutrition and physical activity recommendations should be followed. An exercise program emphasizing walking often improves the sense of well-being, reduces the risk of acute ischemic events, and improves exercise tolerance. Mental health care should be sought when appropriate, particularly for depression and anxiety.

Hypertension (BP > 130/80 for patients with CAD) and elevated blood pressure (systolic BP 120 to 129) are treated diligently because even mild hypertension increases cardiac workload (3). Aggressive reduction of total cholesterol and low-density lipoprotein (LDL) cholesterol (via diet plus statins) slows the progression of CAD, may cause some lesions to regress, and improves endothelial function and thus arterial response to stress. High-intensity statin therapy, when tolerated, is recommended, with ezetimibe, a PSCK9 inhibitor, or other agents as useful adjuncts or alternatives (2, 3). Patients with heart failure and stable angina should be treated per guidelines (5), including receiving a sodium-glucose-cotransporter-2 (SGLT2) inhibitor; patients with type 2 diabetes should receive an SGLT2 inhibitor or a glucagon-like-peptide-1 (GLP-1) receptor agonist.

Antiplatelet agents inhibit platelet aggregation. Aspirin binds irreversibly to platelets and inhibits cyclooxygenase and platelet aggregation. Low-dose aspirin is recommended for all patients with stable angina and chronic coronary artery disease (2, 3). Dual-antiplatelet therapy is used after percutaneous coronary intervention (for stable disease, or after acute coronary syndrome in previously stable disease), with aspirin and a P2Y12 inhibitor (eg, clopidogrel, prasugrel, ticagrelor) usually for 12 months. Patients unable to tolerate one agent should receive the other alone.

Influenza, COVID-19, and pneumococcal vaccination are recommended in patients with chronic coronary artery disease (3).

Colchicine may also be used for secondary prevention in patients with chronic coronary artery disease (3).

Medications for stable angina

In addition to reducing the risk of future events, the main goals of angina treatment are to:

- Relieve symptoms
- Prevent or reduce ischemia

(See also table Medications for Acute Coronary Syndromes and Chronic Coronary Artery Disease.)

To prevent ischemia, several classes of medications are used:

- Beta-blockers: Most patients, unless contraindicated or not tolerated
- Long-acting nitrates: If needed
- Calcium channel blockers: If needed

Beta-blockers are a first-line therapy for patients with stable angina and left ventricular ejection fraction (LVEF) ≤ 50% or previous myocardial infarction; they reduce the risk of death or major adverse cardiovascular event (3). Beta-blockers block sympathetic stimulation of the heart and reduce systolic BP, heart rate, contractility, and cardiac output, thus decreasing myocardial oxygen demand and increasing exercise tolerance. Beta-blockers also increase the threshold for ventricular fibrillation. Most patients tolerate these agents well. Many beta-blockers are available and effective. Dose is titrated upward as needed until limited by bradycardia or adverse effects. Beta-blockers, other than labetalol and carvedilol (6), which also have alpha blocking activity, should not be used in patients with vasospastic angina because they may cause coronary vasospasm from unopposed alpha-receptor activity (1).

Patients who are at risk of beta-blocker intolerance (eg, those with asthma) may be tried on a cardioselective beta-blocker (eg, bisoprolol) perhaps with pulmonary function testing before and after medication administration to detect drug-induced bronchospasm. Patients who cannot tolerate beta-blockers are given a **non-dihydropyridine calcium channel blocker** (eg, diltiazem, verapamil), which has negative chronotropic effects and reduces myocardial oxygen demand (3). Non-dihydropyridine calcium channel blockers can be used alone in patients with beta-blocker intolerance or asthma and normal left ventricular systolic function but may increase cardiovascular mortality in patients with left ventricular systolic dysfunction.

Long-acting nitrates (oral or transdermal) are used if symptoms persist after the beta-blocker dose is maximized, or as a therapy for symptomatic stable angina if a beta-blocker is not indicated or tolerated (3). If angina occurs at predictable times, a nitrate is given to cover those times. Oral nitrates include isosorbide dinitrate and isosorbide mononitrate (the active metabolite of the dinitrate). They are effective within 1 to 2 hours; their effect lasts 4 to 6 hours. Sustained-release formulations of isosorbide mononitrate appear to be effective throughout the day. For transdermal use, cutaneous nitroglycerin patches have largely replaced nitroglycerin ointments, primarily because ointments are inconvenient and messy. Patches slowly release the medication for a prolonged effect; exercise capacity improves 4 hours after patch application and wanes in 18 to 24 hours. Nitrate tolerance may occur, especially when plasma concentrations are kept constant. Because risk of myocardial infarction is highest in early morning, an afternoon or early evening respite period from nitrates is reasonable unless a patient commonly has angina at that time. For nitroglycerin, an 8- to 10-hour respite period seems sufficient. Isosorbide dinitrate requires a 12-hour respite period to minimize the risk of tolerance. If given once a day, sustained-release isosorbide mononitrate does not appear to elicit tolerance.

Dihydropyridine calcium channel blockers may be used if symptoms persist despite use of nitrates or if nitrates are not tolerated (2, 3). Calcium channel blockers are particularly useful if hypertension or coronary spasm is also present. Different types of calcium channel blockers have different effects. Dihydropyridine calcium channel blockers (eg, nifedipine, amlodipine, felodipine) have no chronotropic effects and vary substantially in their negative inotropic effects. Shorter-acting dihydropyridines, particularly nifedipine (7), may cause acute hypotension and reflex tachycardia and are associated with increased mortality in patients with CAD; they should not be used alone to treat stable angina. Longer-acting formulations of dihydropyridines have fewer tachycardic effects; they are most commonly used with a beta-blocker. Among longer-acting dihydropyridines, amlodipine has the weakest negative inotropic effects; it may be used in patients with left ventricular systolic dysfunction (8).

Ranolazine is a **sodium channel blocker** that can be used to treat chronic angina in patients whose symptoms persist despite beta-blockers, long-acting nitrates, or calcium channel blockers (3). Because ranolazine may cause QT interval prolongation, it is usually reserved for patients in whom symptoms persist despite optimal treatment with other antianginal agents. Dizziness, headache, constipation, and nausea may occur (2).

Ivabradine is a **sinus node inhibitor** that inhibits inward sodium/potassium current in a certain gated channel (funny or "f" channel) in sinus node cells, thus slowing heart rate without decreasing contractility. It can be used for symptomatic treatment of chronic stable angina in patients with normal sinus rhythm at 70 beats per minute or greater who cannot take beta-blockers (1) or other anti-anginal therapy. However, it may be harmful in combination with standard therapy (3).

ACE inhibitors, ARBs, or an angiotensin receptor/neprilysin inhibitor are given as indicated for patients with concomitant heart failure, chronic kidney disease, hypertension, or diabetes.

Lipid-lowering agents (statins and others) are generally recommended for all patient with chronic coronary disease (3).

Revascularization

Revascularization, either with PCI (eg, angioplasty and stent placement) or CABG, should be considered if angina persists despite pharmacologic therapy and worsens quality of life or if anatomic lesions (noted during angiography) put a patient at high risk of mortality (3). The choice between PCI and CABG depends on the extent and location of anatomic lesions, the experience of the operator and medical center, and, to some extent, patient preference (9).

PCI is usually preferred for 1- or 2-vessel disease with suitable anatomic lesions and is sometimes used for 3-vessel disease with suitable anatomy. As stent technology improves, PCI is being used for more complicated lesions (eg, lesions that are long or near bifurcation points). PCI can also be used for left main coronary artery disease (LMCA) in the appropriate clinical setting (10, 11, 12, 13). FFR is used to clarify the hemodynamic significance of coronary stenoses when they are equivocal, or in patients who have not had prior evaluation for ischemia, including stress testing (3).

CABG is very effective in selected patients with angina. CABG is superior to PCI in patients with diabetes and multivessel disease, and in some other patients with diffuse, complex multivessel disease that is amenable to grafting (3, 14, 15, 16, 17, 18).

CABG improves survival for patients with left main disease, those with 3-vessel disease and poor left ventricular function, and some patients with 2-vessel disease (18). However, for patients with mild or moderate angina (Canadian Cardiovascular Society class 1 or 2) or 3-vessel disease and good ventricular function, CABG appears to only marginally improve survival. Several studies show better long-term outcomes following CABG than with PCI for patients with diabetes and proximal left anterior descending disease (18). For patients with 1-vessel disease, outcomes with pharmacologic therapy, PCI, and CABG are similar; exceptions are left main disease and proximal left anterior descending disease, for which revascularization (either by PCI or CABG) appears advantageous.

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Prognosis for Stable Angina

The main adverse outcomes of stable angina pectoris are the development of unstable angina, myocardial infarction, or sudden death due to arrhythmias. The annual risk of death or myocardial infarction is approximately 3 to 4% in patients with stable angina receiving medical therapy, no history of MI, a normal resting ECG, and normal BP (1). Factors associated with an increased mortality risk include poor exercise capacity, more severe angina, impaired glucose tolerance or diabetes, decreased glomerular filtration rate, and in males, sexual problems (1, 2).

Prognosis also worsens with increasing age, increasingly severe anginal symptoms, presence of anatomic lesions, and poor ventricular function (3). Lesions in the left main coronary artery or proximal left anterior descending artery indicate particularly high risk. Although prognosis correlates with number and severity of coronary arteries affected, prognosis is surprisingly good for patients with stable angina, even those with 3-vessel disease, if ventricular function is normal.

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Key Points

- **Angina pectoris occurs when cardiac workload exceeds the ability of coronary arteries to supply an adequate amount of oxygenated blood.**
- **Symptoms of stable angina pectoris range from a vague, barely troublesome ache to a severe, intense precordial crushing sensation; they are typically precipitated by exertion, last no more than a few minutes, and subside with rest.**
- **Testing for diagnosis and risk stratification includes electrocardiography, coronary CT angiography, and stress testing.**
- **Give nitroglycerin for immediate relief of angina.**
- **Maintain patients on an antiplatelet agent, a beta-blocker, and a statin, and add a calcium channel blocker for further symptom prevention if needed.**
- **Consider revascularization if significant angina persists despite pharmacologic therapy or if lesions noted during angiography indicate high risk of mortality.**
- **Do coronary angiography when revascularization (percutaneous intervention or coronary artery bypass grafting) is being considered.**
- **Initiate ongoing risk modification in all patients with stable angina, including smoking cessation, blood pressure control, weight control, glycemic control, exercise, and use of statins and antiplatelet agents.**

Drug Information for the Topic

Unstable Angina

(Acute Coronary Insufficiency; Preinfarction Angina; Intermediate Syndrome)

Full Review: Jul 2026 By ***Ranya N. Sweis****, MD, MS, Northwestern University Feinberg School of Medicine |* ***Arif Jivan****, MD, PhD, Northwestern University Feinberg School of Medicine | Peer reviewed by* ***Jonathan G. Howlett****, MD, Cumming School of Medicine, University of Calgary*
Last updated: Jul 2026

Unstable angina is a type of non-ST-segment-elevation acute coronary syndrome (NSTE-ACS) that usually results from acute obstruction of a coronary artery without myocardial infarction. Symptoms include chest discomfort (angina pectoris) with or without dyspnea, nausea, and diaphoresis. Diagnosis is by electrocardiography (ECG) and the absence of a significant increase in cardiac troponin. Immediate treatment is with antiplatelets, anticoagulants, and nitrates, and later, with statins and beta-blockers. Coronary angiography with percutaneous intervention or coronary artery bypass surgery is often necessary. Subsequent care includes lipid management, beta-blockers, cardiac rehabilitation, risk factor management, and dual antiplatelet therapy.

Unstable angina is a type of NSTE-ACS that is defined as temporary myocardial ischemia, leading to decreased blood flow, without myocardial necrosis as evidenced by significant elevation in circulating troponin (1). It is often accompanied by ECG changes and may manifest as:

- Rest angina that is prolonged (usually > 20 minutes)
- New-onset angina of at least class 3 severity in the Canadian Cardiovascular Society (CCS) classification (see table Canadian Cardiovascular Society Classification System for Angina Pectoris)
- Increasing angina, that is, previously diagnosed angina that has become distinctly more frequent, more severe, longer in duration, or lower in threshold (eg, increased by ≥ 1 CCS class or to at least CCS class 3)

Unstable angina carries an increased risk of subsequent myocardial infarction (2).

General references

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Symptoms and Signs of Unstable Angina

Patients have symptoms similar to those of stable angina except that the chest pain or discomfort of unstable angina usually is more intense, lasts longer, occurs spontaneously at rest, or is precipitated by lower degrees of exertion than stable angina, is progressive (crescendo) in nature, or involves any combination of these features.

Unstable angina is classified based on severity and clinical situation (see table Canadian Cardiovascular Classification System for Angina Pectoris). Also considered are whether unstable angina occurs during treatment for chronic stable angina and whether transient changes in ST-T waves occur during angina. If angina has occurred within 48 hours and no contributory extracardiac condition is present, troponin levels may be measured with a highly sensitive assay of cardiac troponin (hs-cTn) to help estimate prognosis; a completely undetectable troponin level indicates a better prognosis than any positive level even if not meeting criteria for NSTEMI (1).

Symptoms and signs reference

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Diagnosis of Unstable Angina

- Serial ECGs
- Serial cardiac troponin measurements
- Sometimes noninvasive risk assessment
- Often coronary angiography

Evaluation begins with initial and serial ECG determinations and serial measurements of cardiac troponin to help distinguish between unstable angina and acute myocardial infarction (MI)—either non–ST-segment elevation MI (NSTEMI) or ST-segment elevation MI (STEMI). This distinction is the center of the decision pathway for several reasons: the diagnosis determines the timing of angiography and percutaneous coronary intervention (PCI) (or other reperfusion strategy), the possibility of a selective noninvasive strategy for unstable angina in patients without certain risk factors, and because treatment with a fibrinolytic is an option for patients with STEMI but not for those with NSTE-ACS.

ECG

ECG is the most important test and should be performed as soon as possible (eg, within 10 minutes of presentation). ECG changes such as ST-segment depression, ST-segment elevation, or T-wave inversion may occur during unstable angina but are transient.

Cardiac troponin

Patients suspected of having an acute coronary syndrome (ACS), including unstable angina should have a highly sensitive assay of cardiac troponin (hs-cTn) performed on presentation and 1 to 2 hours later. If a standard troponin assay is used, measurements are performed at presentation and 3 to 6 hours later. Cardiac troponin, particularly when measured using high-sensitivity troponin tests, may be slightly increased but does not meet the criteria for myocardial infarction (above the 99th percentile of the upper reference limit).

Coronary angiography

Patients with unstable angina without clinical instability (cardiogenic shock, new or worsening heart failure symptoms, refractory ischemic symptoms, hemodynamic or electrical instability) fall into the low-risk NSTE-ACS category by definition (1). Such patients may either undergo coronary angiography before discharge or a noninvasive risk assessment with stress testing or coronary CT angiography. Those with abnormal findings on noninvasive assessment, or those with ECG findings or symptoms of ongoing ischemia, undergo angiography.

Patients presenting with clinical instability undergo immediate angiography.

Coronary angiography most often combines diagnosis with PCI (ie, angioplasty, stent placement).

Diagnosis reference

1. Rao SV, O'Donoghue ML, Ruel M, et al. 2025 ACC/AHA/ACEP/NAEMSP/SCAI Guideline for the Management of Patients With Acute Coronary Syndromes: A Report of the American College of Cardiology/American Heart Association Joint Committee on Clinical Practice Guidelines. *Circulation.* 2025;151(13):e771-e862. doi:10.1161/CIR.0000000000001309

Treatment of Unstable Angina

- Prehospital care: Oxygen, aspirin, and nitrates and triage to an appropriate medical center
- In-hospital pharmacologic therapy: Additional antiplatelet agents, analgesics, anticoagulants, and in some cases other medications
- Sometimes, reperfusion with percutaneous coronary intervention (PCI) or coronary artery bypass grafting (CABG)
- Supportive care
- Post-discharge cardiac rehabilitation and chronic management of coronary artery disease

Prehospital care

- Oxygen
- Aspirin
- Nitrates
- Triage to appropriate medical center

A reliable IV route must be established, oxygen given (typically 2 L by nasal cannula) if patients have hypoxemia, and continuous ECG monitoring started. Prehospital interventions by emergency medical personnel (including ECG, chewed aspirin [160 to 325 mg], pain management with nitroglycerin) can reduce risk of mortality and complications (1). Early diagnostic data and response to treatment can help determine the need for and timing of revascularization and guide triage to the appropriate hospital.

Hospital admission

- Risk-stratify patient to determine need for, timing of, and method of reperfusion (PCI or coronary artery bypass grafting [CABG])
- Pharmacologic therapy with antiplatelets, anticoagulants, and other medications based on reperfusion strategy

On arrival to the emergency department, the patient's diagnosis is confirmed. Pharmacologic therapy and timing of revascularization depend on the clinical picture. In clinically unstable patients (patients with ongoing symptoms, hypotension, or sustained arrhythmias), urgent angiography with revascularization is indicated. In clinically stable patients, a selective invasive approach may be employed (see figure Approach to Unstable Angina).

Approach to pharmacologic therapy is discussed in Medications for Acute Coronary Syndrome, and choice of reperfusion strategy is further discussed in Revascularization for Acute Coronary Syndromes.

Approach to Unstable Angina

a Use morphine or fentanyl judiciously if nitroglycerin is contraindicated or if the patient has symptoms despite nitroglycerin therapy; they may attenuate the effect of P2Y12 inhibitors.

b See Antiplatelet Agents for more detail.

c Based on risk scores, troponin, symptoms, and ECG changes.

d Unstable patients include those with cardiogenic shock, new or worsening heart failure symptoms, refractory ischemic symptoms, and hemodynamic or electrical instability.

e CABG may be preferred to PCI for patients with the following: High complexity coronary disease that involves the left main coronary artery, diabetes and multivessel disease involving

the left anterior descending artery, multivessel or diffuse disease, or severe left ventricular dysfunction with multivessel or complex left main disease.

ACE = angiotensin-converting enzyme inhibitor; ARB = angiotensin II receptor blocker; CABG = coronary artery bypass grafting; CCTA = coronary CT angiography; NSTE–ACS = non–ST-segment elevation–acute coronary syndrome; MINOCA = myocardial infarction with non-obstructive coronary arteries; PCI = percutaneous intervention; SL = sublingual.

Data from Rao SV, O'Donoghue ML, Ruel M, et al. 2025 ACC/AHA/ACEP/NAEMSP/SCAI Guideline for the Management of Patients With Acute Coronary Syndromes: A Report of the American College of Cardiology/American Heart Association Joint Committee on Clinical Practice Guidelines. *Circulation.* 2025;151(13):e771-e862. doi:10.1161/CIR.0000000000001309.

Pharmacologic treatment of unstable angina

All patients should be given antiplatelet agents, anticoagulants, and if chest pain is present, antianginals. The specific medications used depend on the reperfusion strategy and other factors; medication selection and use are discussed in Medications for Acute Coronary Syndromes. Other medications, such as beta-blockers, angiotensin-converting enzyme (ACE) inhibitors or angiotensin II receptor blockers (ARBs), and statins, should be initiated during admission (see Post-ACS Treatment and Rehabilitation and the table Medications for Acute Coronary Syndromes and Chronic Coronary Artery Disease).

Patients with unstable angina should be given the following (unless contraindicated):

- Antiplatelet agents: Aspirin and prasugrel, ticagrelor, or clopidogrel
- Anticoagulants: A heparin (unfractionated or low molecular weight heparin) or bivalirudin
- Sometimes a glycoprotein IIb/IIIa inhibitor when PCI is performed
- Antianginal therapy, usually nitroglycerin
- Beta-blocker
- Angiotensin-converting enzyme (ACE) inhibitor or angiotensin II receptor blocker (ARB)
- Statin and sometimes additional lipid-lowering agents
- Sometimes mineralocorticoid receptor antagonists

All patients are given aspirin 160 to 325 mg (not enteric-coated), if not contraindicated. Chewing the first dose before swallowing quickens absorption. Aspirin reduces short-term and long-term mortality risk (2). In patients undergoing PCI, a loading dose of prasugrel (60 mg orally once), ticagrelor (180 mg orally once), or clopidogrel (300 to 600 mg orally once) is recommended to reduce ischemic events (3). For urgent PCI, prasugrel and ticagrelor are more rapid in onset and may be preferred. IV cangrelor may also be used during PCI to reduce ischemic events in patients who are P2Y12-inhibitor naive (4).

Either a low molecular weight heparin (LMWH), unfractionated heparin, or bivalirudin is given routinely to patients with unstable angina unless contraindicated (eg, by active bleeding) (5). Unfractionated heparin is more complicated to use because it requires frequent (every 6 hours) dosing adjustments to achieve target activated partial thromboplastin time (aPTT). The LMWHs have better bioavailability, are given by simple weight-based dose without monitoring aPTT and dose titration, and have lower risk of heparin-induced thrombocytopenia; they are recommended when an early invasive strategy is not planned. Bivalirudin is recommended for patients with a known or suspected history of heparin-induced thrombocytopenia who are undergoing PCI.

Consider a glycoprotein IIb/IIIa inhibitor during (or sometimes before) PCI for patients with high-risk lesions (eg, high thrombus burden, no reflow). Abciximab, tirofiban, and eptifibatide appear to have equivalent efficacy, and the choice of agent should depend on other factors (eg, cost, availability, familiarity) (35).

Chest pain can be treated with sublingual or intravenous nitroglycerin, or sometimes morphine or fentanyl (5). Nitroglycerin is preferable to opioids, which should be used judiciously (eg, if a patient has a contraindication to nitroglycerin or is in pain despite maximal nitroglycerin therapy). Nitroglycerin is initially given sublingually, followed by continuous IV drip if needed. Morphine, given 2 to 4 mg IV, repeated every 15 minutes as needed, is highly effective but can depress respiration, can reduce myocardial contractility, and is a potent venous vasodilator. Evidence also suggests that morphine and fentanyl interfere with some P2Y12 receptor inhibitor activity (6, 7, 8). A large retrospective trial also showed that morphine may increase mortality in patients with acute myocardial infarction (7, 8). Hypotension and bradycardia may also occur secondary to morphine use, but these complications can usually be overcome by prompt elevation of the lower extremities.

Post-acute pharmacologic treatment

Dual antiplatelet therapy with aspirin and a P2Y12 inhibitor (eg, ticagrelor, clopidogrel) for up to 1 year is recommended (5, 9, 10). A proton pump inhibitor should be given with dual antiplatelet therapy. If PCI was performed, options for tapering to monotherapy sooner than 12 months, with aspirin or a P2Y12 inhibitor, are discussed in more detail under Treatment of Acute Coronary Syndromes and Medications for Acute Coronary Syndromes.

Standard therapy for all patients with acute coronary syndromes includes beta-blockers, ACE inhibitors or ARBs, and statins (5). Beta-blockers are recommended unless contraindicated (eg, by bradycardia, heart block, hypotension, or asthma), especially for high-risk patients. Beta-blockers reduce heart rate, arterial pressure, and contractility, thereby reducing cardiac workload and oxygen demand. However, beta-blockers may have limited benefit in patients with preserved left ventricular ejection fraction (11, 12). ACE inhibitors or ARBs may provide long-term cardioprotection by improving endothelial function (13), particularly in patients with heart failure, hypertension, diabetes, or chronic kidney disease. Statins are also standard therapy regardless of lipid levels and should be continued indefinitely, with the addition of other lipid-lowering medications as needed.

Mineralocorticoid receptor antagonists (spironolactone or eplerenone) are indicated for patients with left ventricular dysfunction and either heart failure symptoms or diabetes following ACS, to reduce morbidity and mortality (5, 14).

Reperfusion therapy in unstable angina

In patients with unstable angina, angiography is often performed prior to discharge if the patient is stable or immediately in unstable patients (with cardiogenic shock, new or worsening heart failure symptoms, refractory ischemic symptoms, hemodynamic or electrical instability). Angiographic findings help determine whether PCI or coronary artery bypass grafting (CABG) is indicated. Choice of reperfusion strategy is further discussed in Revascularization for Acute Coronary Syndromes.

Fibrinolytic drugs, which can be helpful in patients with STEMI without access to immediate angiography and PCI, are not indicated for patients with unstable angina (5).

Pearls & Pitfalls

- Although fibrinolytic drugs can help patients with STEMI, they are not indicated in unstable angina.

Rehabilitation and post-discharge treatment

- Functional evaluation
- Cardiac rehabilitation
- Changes in lifestyle: Regular exercise, diet modification, weight loss, smoking cessation

Post-acute pharmacologic therapy is discussed above.

Patients who did not have coronary angiography during admission, have no high-risk features (eg, heart failure, recurrent angina, ventricular tachycardia or ventricular fibrillation after 24 hours, mechanical complications such as new murmurs, shock), and have an ejection fraction > 40% usually should have stress testing of some sort before or shortly after discharge.

Supervised cardiac rehabilitation programs based in clinics or hospitals or home-based programs are recommended for all patients after acute coronary syndromes (5) and decrease cardiovascular mortality after revascularization (15).

The acute illness and treatment of unstable angina should serve as a catalyst for discussion of modifiable cardiovascular risk factors. Evaluating the patient's physical and emotional status and discussing them with the patient, advising about lifestyle (eg, smoking, diet, work and play habits, exercise), and aggressively managing risk factors may improve prognosis.

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Prognosis for Unstable Angina

Prognosis after an episode of unstable angina depends upon the extent and number of coronary arteries involved as well as other factors such as age, symptom severity, degree of ST-segment deviation, and pre-existing risk of coronary artery disease (1, 2). In addition to estimating prognosis, calculation of risk using the TIMI or GRACE scores can guide the decision to revascularize versus taking a more conservative approach in unstable angina (3).

Studies using the current definition of unstable angina with a high-sensitivity cardiac troponin assay report 1-year myocardial infarction rate of 1.4 to 1.9% and a 1-year overall mortality rate of 0.6 to 2.2% (4, 5).

CLINICAL CALCULATORS

Thrombolysis in Myocardial Infarction (TIMI) Score for Unstable Angina Non ST Elevation Myocardial Infarction

CLINICAL CALCULATORS

GRACE 2.0 Score for Acute Coronary Syndrome Prognosis (non-linear model 2014)

Prognosis references

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Key Points

- **Unstable angina is defined as myocardial ischemia associated with reduced coronary flow, but without myonecrosis.**
- **Symptoms of unstable angina include new or worsening chest pain or chest pain occurring at rest.**

- Diagnosis is based on symptoms, serial ECG findings and the absence of a significant elevation in cardiac troponin.
- Immediate treatment includes oxygen, antianginals, antiplatelets, and anticoagulants.
- For patients with refractory symptoms, shock, hypotension, or sustained arrhythmias, do immediate angiography.
- For stable patients, do angiography during hospitalization, or consider non-invasive risk stratification.
- Following recovery, initiate or continue antiplatelet agents, beta-blockers, ACE inhibitors or angiotensin II receptor blockers, and statins.
- Initiate cardiac rehabilitation and aggressively manage risk factors.

Drug Information for the Topic

MERCK MANUAL
Professional Version

Overview of Acute Coronary Syndromes (ACS)

(Unstable Angina; Acute MI; Myocardial Infarction)

*Full Review: Jul 2026 By **Ranya N. Sweis**, MD, MS, Northwestern University Feinberg School of Medicine | **Arif Jivan**, MD, PhD, Northwestern University Feinberg School of Medicine | Peer reviewed by **Jonathan G. Howlett**, MD, Cumming School of Medicine, University of Calgary*
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Acute coronary syndromes usually result from acute obstruction of a coronary artery. Consequences depend on degree and location of obstruction and range from unstable angina to non–ST-segment elevation myocardial infarction (NSTEMI), to ST-segment elevation myocardial infarction (STEMI), sometimes leading to sudden cardiac death. Symptoms are similar in each of these syndromes and include chest discomfort with or without dyspnea, nausea, and diaphoresis. Diagnosis is by electrocardiography (ECG) and biomarkers. Immediate treatment is with antiplatelets, anticoagulants, nitrates and, for STEMI and intermediate- and high-risk NSTEMI, revascularization via percutaneous intervention, fibrinolytic medications (for STEMI only), or occasionally, coronary artery bypass graft surgery. Subsequent care includes lipid management, beta-blockers, cardiac rehabilitation, risk factor management, and dual antiplatelet therapy.

Classification of Acute Coronary Syndromes

Acute coronary syndromes (ACS) include (1):

- Unstable angina
- Non–ST-segment elevation myocardial infarction (NSTEMI)
- ST-segment elevation myocardial infarction (STEMI)

These syndromes all involve acute coronary ischemia and are distinguished based on symptoms, ECG findings, and cardiac troponin levels. It is helpful to distinguish the syndromes because prognosis and treatment vary. Unstable angina and NSTEMI are also collectively referred to as non-ST elevation ACS (NSTE-ACS).

Unstable angina (historically referred to as acute coronary insufficiency, preinfarction angina, or intermediate syndrome) is defined as transient myocardial ischemia with reduced coronary blood flow, in the absence myocardial necrosis (circulating cardiac troponin < 99th percentile).

Symptoms of unstable angina include:

- Rest angina that is prolonged (usually > 20 minutes)
- New-onset angina of at least class 3 severity in the Canadian Cardiovascular Society (CCS) classification (see table Canadian Cardiovascular Society Classification System for Angina Pectoris)
- Increasing angina, that is, previously diagnosed angina that has become distinctly more frequent, more severe, longer in duration, or lower in threshold (eg, increased by ≥ 1 CCS class or to at least CCS class 3)

However, these symptoms occur across the range of acute coronary syndromes, which are differentiated based on ECG changes and cardiac troponin levels.

ECG changes such as ST-segment depression, ST-segment elevation, or T-wave inversion may occur during **unstable angina**, but they are transient. Troponin may be detectable with high-sensitivity troponin tests (hs-cTn), but, by definition, is < 99th percentile of the upper reference limit (URL) (1, 2). Unstable angina is, as the name suggests, clinically unstable and often a prelude to myocardial infarction or arrhythmias or, less commonly, to sudden death.

Non–ST-segment elevation MI (NSTEMI, subendocardial MI) is myocardial necrosis (evidenced by circulating cardiac troponin levels ≥ 99th percentile) without acute ST-segment elevation that meets criteria for STEMI. ECG changes such as ST-segment depression, T-wave inversion, or both may be present. NSTEMI is characterized by ischemia that is ongoing rather than transient as in unstable angina, and myocardial necrosis that is typically, though not universally, subendocardial.

ST-segment elevation MI (STEMI, transmural MI) is myocardial necrosis with ECG changes showing ST-segment elevation that is not quickly reversed by nitroglycerin. The presence of a new left bundle branch block is not considered a STEMI equivalent in isolation (1, 3). Cardiac troponin is elevated, and myocardial necrosis is transmural.

Both types of MI may or may not produce Q waves on the ECG.

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Etiology of Acute Coronary Syndromes

The **most common cause** of acute coronary syndromes is (1):

- Disruption of an unstable atherosclerotic plaque with thrombosis in the atherosclerotic coronary artery

Atheromatous plaque sometimes becomes unstable or inflamed, causing it to erode or rupture, exposing thrombogenic material, which activates platelets and the coagulation cascade and produces an acute thrombus. Platelet activation involves a conformational change in membrane glycoprotein (GP) IIb/IIIa receptors, allowing cross-linking (and thus aggregation) of platelets. The resultant thrombus abruptly interferes with blood flow to parts of the myocardium.

Even atheromas causing minimal obstruction can rupture and result in thrombosis; in one large prospective imaging trial, the majority of patients having myocardial infarction or cardiovascular death did not have evidence of significant coronary stenosis at baseline (2, 3). Thus, although the severity of stenosis helps predict symptoms, it does not always predict acute thrombotic events.

Spontaneous thrombolysis occurs in up to one-quarter of patients (4); 24 hours later, thrombotic obstruction is found in only approximately one-third (5). However, in virtually all cases, obstruction lasts long enough to cause varying degrees of tissue necrosis.

Myocardial infarction in the absence of obstructive coronary artery disease (MINOCA)

Myocardial infarction with nonobstructive coronary arteries (MINOCA) is found in approximately 5 to 6% of patients with acute MI who undergo coronary angiography (6). Patients with MINOCA tend to be younger, female, and without dyslipidemia, and they have myocardial necrosis without significant coronary atherosclerosis.

Causes of MINOCA include:

- Coronary artery embolism
- Coronary spasm
- Spontaneous coronary artery dissection

Coronary arterial embolism (as distinct from the typical atherothrombosis by having a non-coronary source and the possibility of other non-thrombotic material such as air) can occur in mitral stenosis, aortic stenosis, infective endocarditis, nonbacterial thrombotic endocarditis (marantic endocarditis), or atrial fibrillation.

Cocaine use and other causes of coronary spasm can sometimes result in myocardial infarction. Spasm-induced MI may occur in normal or atherosclerotic coronary arteries.

Spontaneous coronary artery dissection is a non-traumatic tear in the coronary intima with creation of a false lumen. Blood flowing through the false lumen expands it, which restricts blood flow through the true lumen, sometimes causing coronary ischemia or infarction. Dissection may occur in atherosclerotic or non-atherosclerotic coronary arteries. Non-atherosclerotic dissection is more likely in pregnant or postpartum patients and/or patients with fibromuscular dysplasia or other connective tissue disorders.

Plaque disruption in atherosclerotic coronary arteries without significant narrowing can also cause MINOCA.

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Pathophysiology of Acute Coronary Syndromes

Initial consequences vary with size, location, and duration of obstruction and range from transient ischemia to infarction. Measurement of troponin with a high sensitivity test indicates that some cell necrosis probably occurs even with mild ischemia; thus, ischemic events occur on a continuum, and classification into subgroups, although useful, is somewhat arbitrary. Sequelae of the acute event depend primarily on the mass and type of cardiac tissue infarcted.

Myocardial dysfunction

Ischemic (but not infarcted) tissue has impaired contractility and relaxation, resulting in hypokinetic or akinetic segments; these segments may expand or bulge during systole (called paradoxical motion). The size of the affected area determines the clinical effect, which ranges from minimal or mild heart failure to cardiogenic shock; usually, large parts of myocardium must be ischemic to cause significant myocardial dysfunction. Some degree of heart failure occurs in up to approximately one-third of patients hospitalized with acute myocardial infarction (1, 2). It is termed ischemic cardiomyopathy if low cardiac output and heart failure persist. Ischemia involving the papillary muscle may lead to mitral valve regurgitation. Dysfunctional wall motion can allow mural thrombus formation.

Myocardial infarction (MI)

Myocardial infarction is myocardial necrosis resulting from abrupt reduction in coronary blood flow to part of the myocardium. Infarcted tissue is permanently dysfunctional; however, there is a zone of potentially reversible ischemia adjacent to infarcted tissue. MI affects predominantly the left ventricle (LV), but damage may extend into the right ventricle (RV) or the atria.

Infarction may be:

- Transmural: Transmural infarcts involve the whole thickness of myocardium from epicardium to endocardium and are usually characterized by abnormal Q waves on ECG.
- Nontransmural (subendocardial): Nontransmural infarcts do not extend through the ventricular wall and cause only ST-segment and T-wave (ST-T) abnormalities on ECG.

Because the transmural depth of necrosis cannot be precisely determined clinically, infarcts are classified as STEMI or NSTEMI by the presence or absence of ST-segment elevation, acutely, on the ECG.

Necrosis of a significant portion of the interventricular septum or ventricular wall may rupture, with dire consequences. A ventricular aneurysm or pseudoaneurysm may form.

Electrical dysfunction

Electrical dysfunction can be significant in any form of acute coronary syndrome. Ischemic and necrotic cells are incapable of normal electrical activity, resulting in various ECG changes (predominantly ST-T abnormalities), arrhythmias, and conduction disturbances. ST-T abnormalities of ischemia include ST-segment depression (often downsloping from the J point), T-wave inversion, ST-segment elevation (often referred to as injury current), and peaked T waves in the hyperacute phase of infarction. Conduction disturbances can reflect damage to the sinus node, the atrioventricular (AV) node, or specialized conduction tissues. Most changes are transient; some are permanent.

Types of ST-Segment Depression

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Symptoms and Signs of Acute Coronary Syndromes

Symptoms of acute coronary syndromes depend somewhat on the extent and location of obstruction and are quite variable. Painful stimuli from thoracic organs, including the heart, can cause discomfort described as pressure, tearing, gas with the urge to eructate, indigestion, burning, aching, stabbing, and sometimes sharp needle-like pain. Many patients describe their symptoms as discomfort rather than pain. Except when infarction is massive, recognizing the amount of ischemia by symptoms alone is difficult.

Symptoms of ACS are similar to those of angina and are discussed in more detail in sections on unstable angina and acute myocardial infarction. Of note, some patients with coronary artery disease (particularly patients with diabetes) have silent myocardial ischemia, which is typically detected during stress testing.

Complications

After the acute event, many complications can occur. They may involve:

- Electrical dysfunction (eg, conduction defects, arrhythmias, sudden cardiac death)
- Myocardial dysfunction (eg, heart failure, interventricular septum or free wall rupture, ventricular aneurysm, pseudoaneurysm, mural thrombus formation, cardiogenic shock)
- Valvular dysfunction (typically mitral regurgitation)

Electrical dysfunction can be significant in any form of ACS, but usually, large parts of myocardium must be ischemic to cause significant myocardial dysfunction. Other complications of ACS include recurrent ischemia, left ventricular thrombus, and pericarditis. Pericarditis can occur during or immediately after MI or more than one week later (post-MI syndrome, or Dressler syndrome).

Diagnosis of Acute Coronary Syndromes

- Early and serial ECGs
- Serial cardiac troponin
- Immediate coronary angiography for patients with STEMI or NSTE-ACS with complications
- Delayed angiography for patients with high- or intermediate-risk NSTE-ACS)
- Angiography or noninvasive risk stratification for patients with low-risk NSTE-ACS

Acute coronary syndromes should be considered in males, usually those > age 30 years, and females usually > age 40 years (younger in patients with diabetes), whose main symptom is chest pain or discomfort. Pain must be differentiated from the pain of disorders such as pneumonia, pulmonary embolism, pericarditis, rib fracture, costochondral separation, esophageal spasm, acute aortic dissection, renal calculus, splenic infarction, or various abdominal disorders. In patients with previously diagnosed hiatus hernia, peptic ulcer, or a gallbladder disorder, the clinician must be wary of attributing new symptoms to these disorders. (For approach to diagnosis, see also Chest Pain.)

The initial diagnostic approach is the same when any ACS is suspected: initial and serial ECGs and serial cardiac biomarker measurements, which distinguish among unstable angina, NSTEMI, and STEMI. Every emergency department should have a triage system to immediately identify patients with chest pain for rapid assessment and ECG. Pulse oximetry and chest radiograph (particularly to look for mediastinal widening, which suggests aortic dissection) are also performed.

ECG

ECG is the most important test and should be performed as soon as possible (within 10 minutes of presentation to medical care, whether in or out of the hospital) (1). It is the center of the decision pathway because emergent cardiac catheterization is indicated for patients with acute STEMI but not normally for those with NSTEMI. In addition, fibrinolytics benefit patients with STEMI but may increase risk for those with NSTEMI.

STEMI

For STEMI, initial ECG is usually diagnostic, showing ST-segment elevation ≥ 1 mm in 2 or more contiguous leads subtending the damaged area (see figure Acute Lateral Left Ventricular Infarction) (1).

In leads V2-V3 the elevation must be 1.5 to 2.5 mm depending on age and gender.

Leads in addition to the standard 12 may be used to better detect right ventricular (RV) (V4R) or posterior (V7-V9) infarction.

Acute Lateral Left Ventricular Infarction (Tracing Obtained Within a Few Hours of Illness Onset)

There is striking hyperacute ST-segment elevation in leads I, aVL, V_4, and V_6 and reciprocal depression in other leads.

The ECG must be read carefully because ST-segment elevation may be subtle, particularly in the inferior leads (II, III, aVF); sometimes the reader's attention is mistakenly focused on leads with ST-segment depression. If symptoms are characteristic, ST-segment elevation on ECG has a reported specificity of 94 to 98% and a sensitivity of 55 to 56% for diagnosing MI (2, 3). Serial tracings (obtained every 8 hours for 1 day, then daily) showing a gradual evolution toward a stable, more normal pattern or development of abnormal Q waves over a few days (see figure Inferior (Diaphragmatic) Left Ventricular Infarction) tend to confirm the diagnosis. However, pathologic Q waves are not necessary for the diagnosis of STEMI or transmural infarction.

Inferior (Diaphragmatic) Left Ventricular Infarction (After the First 24 Hours)

Significant Q waves develop with decreasing ST-segment elevation in leads II, III, and aVF.

NSTEMI

Because nontransmural infarcts are usually in the subendocardial or midmyocardial layers, they do not produce diagnostic ST-segment elevation (or usually Q waves) on the ECG. Instead, they commonly produce only varying degrees of ST-T abnormalities that are less striking, variable, or nonspecific and sometimes difficult to interpret (NSTEMI).

Typical ECG findings in NSTEMI include:

- Horizontal or downsloping ST depression

- T wave inversion
- Transient ST elevation

Non–ST-Segment Elevation Myocardial Infarction (NSTEMI) ECG Findings

IMAGE

Panels A, B, and D show ST depression; panel C shows T wave inversion.

凌鲲 宋/STOCK.ADOBE.COM

If such abnormalities resolve (or worsen) on repeat ECGs, ischemia is very likely. However, when repeat ECGs are unchanged, acute MI is unlikely and, if still suspected clinically, requires other evidence to make the diagnosis. A normal ECG taken when a patient is free of pain does not exclude unstable angina; a normal ECG taken when a patient is experiencing pain, although it does not exclude angina, suggests that the pain is not ischemic.

Cardiac troponin

Cardiac troponins (cTn, referring to subunits T and I, which are specific to cardiac muscle) are the preferred biomarkers used in the diagnosis of acute coronary syndromes. An abnormal cTn level, using an appropriate cutoff, is the primary distinguishing factor between unstable angina and myocardial infarction. Highly sensitive cTn assays are preferred (1).

Troponin is released into the bloodstream after myocardial cell necrosis and, to a lesser extent, during stress and ischemia.

Standard cardiac troponin (cTn) assays, which have been in use for many years, are sensitive and specific, but they are unlikely to detect cardiac troponins except in patients who have an acute cardiac disorder such as a myocardial infarction. Thus, a "positive" cTn test (ie, above the limit of detection) is very specific.

Highly sensitive assays of cardiac troponin (hs-cTn) can detect small amounts of troponin in many healthy people. Thus, troponin levels detected with hs-cTn tests need to be referenced to the normal range and are defined as "elevated" only when higher than 99% of the reference population (1). Furthermore, although an elevated troponin level indicates myocardial cell injury, it does not indicate the cause of the damage (although any troponin elevation increases the risk of adverse outcomes in many disorders) (4). In addition to acute coronary syndromes, many other cardiac and non-cardiac disorders can elevate cardiac troponin levels (see table Causes of Elevated Troponin Levels); not all elevated levels detected with hs-cTn assays represent myocardial infarction, and not all myocardial necrosis results from an acute coronary syndrome event, even when the etiology is ischemic. However, by detecting lower levels of troponin, hs-cTn assays enable earlier identification of MI than other assays, and they have replaced other cardiac biomarker tests in guidelines and in practical use in many centers.

For troponin T, an abnormal (> 99th percentile cutoff) highly sensitive assay has a sensitivity of 95% and a specificity of 80% for detecting myocardial infarction, compared with a sensitivity of 83% and specificity of 93% for the standard assay (4). For troponin I, the highly sensitive assay has a sensitivity of 82% and a specificity of 92%, compared with a sensitivity of 79% and specificity of 95% for the standard assay.

Patients suspected of having an acute coronary syndrome should have an hs-cTn assay performed on presentation and again 1 to 2 hours later. Troponin should be measured at 0 and 3 to 6 hours if a standard cTn assay is used. Because troponin elevation lags behind the onset of myocardial injury, angiography and treatment in patients with STEMI, or unstable patients with a NSTE-ACS, should not be delayed because of an initially normal troponin level.

An hs-cTn level must be interpreted based on the patient's pre-test probability of disease, which is estimated clinically based on:

- Risk factors for ACS
- Symptoms
- ECG findings

A high pre-test probability plus an elevated troponin level detected with an hs-cTn assay is highly suggestive of ACS, whereas a low pre-test probability plus a normal hs-cTn assay result is unlikely to represent ACS (5). Diagnosis is more challenging when test results are discordant with pre-test probability, in which case serial hs-cTn assays often help. A patient with low pre-test probability and an initially slightly elevated troponin level detected with a hs-cTn assay that remains stable on repeat testing probably has non-ACS cardiac disease (eg, heart failure, stable coronary artery disease). However, if the repeat level rises significantly (ie, > 20 to 50%), the likelihood of ACS becomes much higher. If a patient with high pre-test probability has a normal troponin level detected with a hs-cTn assay and that rises > 50% on repeat testing, ACS is likely; but continued normal levels (often including at 6 hours and beyond when suspicion is high) suggest need to pursue an alternate diagnosis.

TABLE

Causes of Elevated Troponin Levels

Type	Conditions

Type	Conditions
Myocardial ischemia	
ACS	Classic AMI • STEMI • NSTEMI
Non-ACS (coronary)	Increased demand (stable coronary artery disease lesion) Coronary artery spasm, embolism, or dissection Procedure-related (PCI, CABG) Cocaine or methamphetamine
Non-ACS (noncoronary)	Hypoxia Global ischemia Hypoperfusion Cardiothoracic surgery
Sudden cardiac death	—
Direct myocardial damage or stress (primarily nonischemic)	
Cardiac disorders	Heart failure Cardiomyopathy (eg, hypertrophic, viral) Hypertension Myocarditis Pericarditis Tachyarrhythmias Injury (ablation procedures, cardiac contusion, cardioversion, electrical shock) Cancer Infiltrative disorders (eg, amyloidosis)
Systemic disorders	Pulmonary embolism Toxicity (eg, anthracyclines) Trauma (eg, severe burns)

Type	Conditions
	Extreme exertion Renal failure Sepsis Stroke Subarachnoid hemorrhage
Analytical	
Assay-based	Poor performance Calibration errors
Sample-based	Heterophile antibody Interference by substances

ACS = acute coronary syndrome; AMI = acute MI; CABG = coronary artery bypass grafting; MI = myocardial infarction; NSTEMI = non–ST-segment elevation MI; PCI = percutaneous coronary intervention; STEMI = ST-segment elevation MI.

Coronary angiography

Coronary angiography most often combines diagnosis with percutaneous coronary intervention (PCI—ie, angioplasty, stent placement). When possible, emergency coronary angiography and PCI are performed as soon as possible after the onset of acute myocardial infarction (primary PCI). In many studies, a shorter interval to PCI ("door to balloon" or "onset to balloon" time) is associated with significantly lower morbidity and mortality and improved long-term outcomes (6, 7, 8, 9).

Invasive angiography is obtained urgently for patients with STEMI and for those with cardiogenic shock, new or worsening heart failure, chest pain refractory to treatment, or hemodynamic instability (including due to arrhythmia). Patients with high- or intermediate-risk NSTEMI typically undergo angiography within the first 24 to 72 hours of hospitalization to detect lesions that may require treatment; those with low-risk NSTE-ACS, including unstable angina, may undergo angiography or noninvasive testing such as stress testing or coronary CT angiography (see Risk Stratification in NSTE-ACS) (1).

Other tests

Routine laboratory tests are nondiagnostic but, if obtained, show nonspecific abnormalities compatible with tissue necrosis (eg, increased erythrocyte sedimentation rate, moderately elevated white blood cell count with a shift to the left). A fasting lipid profile should be obtained as soon as possible, and preferably within the first 24 hours (before low-density lipoprotein [LDL] levels begin to decrease), for all patients hospitalized with ACS.

Myocardial imaging is not needed to make the diagnosis if cardiac biomarkers or ECG are positive. However, in patients with myocardial infarction, bedside echocardiography is invaluable for detecting mechanical complications. Before or shortly after discharge, patients with symptoms suggesting an ACS but with nondiagnostic ECGs and normal cardiac biomarker levels, in whom angiography is not performed should undergo a stress imaging test (radionuclide or echocardiographic imaging with pharmacologic or exercise stress) or coronary CT angiography (1). Imaging abnormalities in such patients indicate increased risk of complications in the next 3 to 6 months and suggest need for angiography (10), which should be performed before discharge or soon thereafter, with revascularization as necessary.

Right heart catheterization using a balloon-tipped pulmonary artery catheter can be used to measure right heart, pulmonary artery, and pulmonary artery occlusion pressures and cardiac output. This procedure is not routinely recommended and should be performed only if patients have significant complications (eg, severe heart failure, hypoxia, hypotension) and by clinicians experienced with catheter placement and management protocols.

Risk stratification and management strategy for NSTE-ACS

For patients with NSTE-ACS, risk scores (Thrombolysis in Myocardial Infarction [TIMI] or the Global Registry of Acute Coronary Events [GRACE]) (11, 12) can be used in combination with other clinical features to determine the timing of angiography versus a noninvasive initial approach. As with any decision pathway, individual patient factors must be taken into account.

CLINICAL CALCULATORS

GRACE 2.0 Score for Acute Coronary Syndrome Prognosis (non-linear model 2014)

For NSTE-ACS patients with cardiogenic shock, new or worsening heart failure, chest pain refractory to treatment, or hemodynamic instability (including due to arrhythmia), coronary angiography with the intent to revascularize is indicated urgently (within 2 hours) (see table Management Strategy for NSTE-ACS).

For NSTE-ACS patients with a GRACE score > 140, steeply rising troponin values, or ongoing dynamic ST-segment changes, coronary angiography with the intent to revascularize is recommended within 24 hours.

For NSTE-ACS patients with a GRACE score 109 to 140, no ongoing ischemic symptoms, and stable (or down trending) troponin levels, coronary angiography with the intent to revascularize is recommended before discharge or within 72 hours.

For NSTE-ACS patients with a GRACE score < 109 and TIMI score < 2, with no ongoing ischemic changes, a troponin level < 99th percentile, and no dynamic ST changes, coronary angiography may be performed, or a noninvasive approach with a stress test or coronary CT angiogram may be used.

TABLE

Management Strategy for NSTE-ACS

Risk Classification	Clinical Features	Management Strategy
Low risk	Any of: • GRACE risk score < 109 • TIMI risk score < 2 • No ongoing symptoms of ischemia • Tn < 99th percentile (ie, unstable angina) • No dynamic ST-segment changes	Routine invasive: Coronary angiography before hospital discharge OR Selective invasive: Initial noninvasive risk stratification stress test or coronary CT angiogram, followed by coronary angiography if indicated
Intermediate risk	Any of: • GRACE risk core 109–140 • No ongoing symptoms of ischemia • Stable or decreasing Tn values	Routine invasive: Coronary angiography before hospital discharge (< 72 hours)
High risk	Any of: • GRACE risk score > 140 • Steeply increasing Tn values on serial testing despite optimal pharmacologic treatment • Ongoing dynamic ST-segment changes	Routine invasive: Coronary angiography < 24 hours
Very high risk (unstable patient)	Any of: • Cardiogenic shock • Symptoms of heart failure (eg, new/worsening mitral regurgitation, acute pulmonary edema)	Immediate invasive: Coronary angiography with intent to revascularize < 2 hours

Risk Classification	Clinical Features	Management Strategy
	• Refractory angina • Hemodynamic or electrical instability (eg, sustained VT, VF)	

GRACE = Global Registry of Acute Coronary Events; NSTE-ACS = non-ST elevation acute coronary syndrome; TIMI = Thrombolysis in Myocardial Infarction; Tn = cardiac troponin; VF = ventricular fibrillation; VT = ventricular tachycardia

Rao SV, O'Donoghue ML, Ruel M, et al. 2025 ACC/AHA/ACEP/NAEMSP/SCAI Guideline for the Management of Patients With Acute Coronary Syndromes: A Report of the American College of Cardiology/American Heart Association Joint Committee on Clinical Practice Guidelines. Circulation. 2025;151(13):e771-e862. doi:10.1161/CIR.0000000000001309

CLINICAL CALCULATORS

CLINICAL CALCULATORS

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Treatment of Acute Coronary Syndromes

- Prehospital care: Oxygen, aspirin, and nitrates and triage to an appropriate medical center
- In-hospital pharmacologic therapy: Additional antiplatelet agents, analgesics, anticoagulants, and in some cases other medications
- Often, reperfusion with percutaneous coronary intervention (sometimes with fibrinolytics, or coronary artery bypass grafting)
- Supportive care, including management of shock and hemodynamic or electrical instability
- Post-discharge cardiac rehabilitation and chronic management of coronary artery disease

Treatment, including pharmacologic therapy, is designed to relieve distress, interrupt thrombosis, reverse ischemia, limit infarct size, reduce cardiac workload, and prevent and treat complications. An acute coronary syndrome is a medical emergency; outcome is greatly influenced by rapid diagnosis and treatment.

Treatment occurs simultaneously with diagnosis.

Contributing disorders (eg, anemia, heart failure) are aggressively treated.

Because the chest pain of myocardial infarction usually subsides within 12 to 24 hours, any chest pain that remains or recurs later is investigated. It may indicate such complications as recurrent ischemia, pericarditis, pulmonary embolism, pneumonia, gastritis, or ulcer.

Prehospital care

- Oxygen
- Aspirin
- Nitrates
- Triage to appropriate medical center

A reliable IV route must be established, oxygen given (typically 2 L by nasal cannula) if patients are hypoxemic (oxygen saturation < 90%), and continuous ECG monitoring started. Prehospital interventions by emergency medical personnel—including ECG, chewed aspirin (160 to 325 mg), pain management with nitrates (see Medications for Acute Coronary Syndromes), triage to the appropriate hospital where timely primary percutaneous coronary intervention (PCI) is available, or early thrombolysis when primary PCI is unavailable—can reduce risk of mortality and complications (1, 2).

Although opioids have long been used to treat pain in patients with acute coronary syndromes, data suggest that morphine and fentanyl attenuate activity of some P2Y12 receptor inhibitors and may contribute to worse patient outcomes (3, 4, 5).

Early electrocardiography, data from other diagnostic testing, and response to treatment can help determine the need for and timing of revascularization. Patients with ACS, particularly STEMI, are preferentially brought by emergency medical services to primary PCI-capable centers.

Hospital admission

- Risk-stratify patient and choose a reperfusion strategy (PCI, fibrinolytics, or coronary artery bypass grafting [CABG] for patients with STEMI and PCI or CABG for patients with unstable angina or NSTE-ACS)
- Pharmacologic therapy with antiplatelets, anticoagulants, and other medications based on reperfusion strategy

On arrival to the emergency department, the patient's diagnosis is confirmed (in some centers patients with STEMI may be transferred directly to the cardiac catheterization laboratory for initial assessment). Pharmacologic therapy and choice of revascularization depend on the type of acute coronary syndrome as well as the clinical picture (see figure Approach to Acute Coronary Syndromes , Pharmacologic Treatment for Unstable Angina and Pharmacologic Treatment of Myocardial Infarction, and in more detail at Medications for Acute Coronary Syndrome, and choice of reperfusion strategy is further discussed in Revascularization for Acute Coronary Syndromes).

Approach to Acute Coronary Syndromes

a Use morphine or fentanyl judiciously if nitroglycerin is contraindicated or if the patient has symptoms despite nitroglycerin therapy; they may attenuate the effect of P2Y12 inhibitors.

b See Antiplatelet Agents for more detail.

c Based on risk scores, troponin, symptoms, and ECG changes.

d Unstable patients include those with cardiogenic shock, new or worsening heart failure symptoms, refractory ischemic symptoms, and hemodynamic or electrical instability.

e 90 minutes if patient is in the United States and calls 911 or presents to a PCI-capable hospital; 120 minutes with hospital-to-hospital transfer.

f CABG may be preferred to PCI for patients with the following: High complexity coronary disease that involves the left main coronary artery, diabetes and multivessel disease involving

the left anterior descending artery, multivessel or diffuse disease, or severe left ventricular dysfunction with multivessel or complex left main disease.

ACE = angiotensin-converting enzyme inhibitor; ARB = angiotensin II receptor blocker; CABG = coronary artery bypass grafting; CCTA = coronary CT angiography; NSTE–ACS = non–ST-segment elevation–acute coronary syndrome; MINOCA = myocardial infarction with non-obstructive coronary arteries; PCI = percutaneous intervention; SL = sublingual; STEMI = ST-segment elevation myocardial infarction.

Data from Rao SV, O'Donoghue ML, Ruel M, et al. 2025 ACC/AHA/ACEP/NAEMSP/SCAI Guideline for the Management of Patients With Acute Coronary Syndromes: A Report of the American College of Cardiology/American Heart Association Joint Committee on Clinical Practice Guidelines. *Circulation.* 2025;151(13):e771-e862. doi:10.1161/CIR.0000000000001309.

When the diagnosis is unclear, bedside cardiac biomarker measurements can help identify low-risk patients with a suspected ACS (eg, those with initially negative cardiac biomarkers and nondiagnostic ECGs), who can be managed in 24-hour observation units or chest pain centers. Higher-risk patients should be admitted to a monitored inpatient unit or critical care unit (CCU). Several validated tools can help stratify risk. Thrombolysis in MI (TIMI) risk scores may be the most widely used (6, 7). For more detail, see Risk stratification and management strategy for NSTE-ACS.

Patients with suspected NSTEMI and intermediate or high risk should be admitted to an inpatient care unit or CCU. Those with STEMI or who are unstable should be admitted to a CCU.

Only heart rate and rhythm recorded by single-lead ECG are consistently useful for routine, continuous monitoring. However, some clinicians recommend routine multilead monitoring with continuous ST-segment recording to identify transient, recurrent ST-segment elevations or depressions. Such findings, even in patients without symptoms, suggest ischemia and identify patients at higher risk who may require more aggressive evaluation and treatment.

Qualified nurses can interpret the ECG for arrhythmia and initiate protocols for its treatment. All staff members should know how to do cardiopulmonary resuscitation (CPR).

Supportive care

If patients present in cardiogenic shock, revascularization of the primary involved vessel, by PCI or CABG, is performed (8). Mechanical circulatory support with a microvascular intra-axial flow device to directly unload the left ventricle can be used in patients with refractory or severe shock. Other hemodynamic and rhythm-related issues are addressed.

Patients with successful, uncomplicated primary PCI for acute MI may be ambulated quickly and be safely discharged in 2 to 4 days.

If reperfusion is not successful or complications are present, patients require longer monitoring for hemodynamic and electric instability while also initiating ambulation. Prolonged bed rest results in rapid physical deconditioning, with development of orthostatic hypotension, decreased work capacity, increased heart rate during exertion, and increased risk of deep venous thrombosis. Prolonged bed rest also intensifies feelings of depression and helplessness.

Anxiety, mood changes, and denial are common. A mild tranquilizer (usually a benzodiazepine) is often given, but many experts believe such medications are rarely needed. Reactive depression is common by the third day of illness and is almost universal at some time during recovery.

After the acute phase of illness, the most important tasks are often management of depression, rehabilitation, and institution of long-term preventive programs. Overemphasis on bed rest, inactivity, and the seriousness of the disorder reinforces anxiety and depressive tendencies, so patients are encouraged to sit up, get out of bed, and engage in appropriate activities as soon as possible. The effects of the disorder, prognosis, and individualized rehabilitation program should be explained to the patient.

Maintaining normal bowel function with stool softeners (eg, docusate) to prevent straining is important. Urinary retention is common among older patients, especially after several days of bed rest or if atropine was given. A catheter may be required but can usually be removed when the patient can stand or sit to void.

For patients who smoke, smoking cessation should be addressed during the hospitalization. All caregivers should devote considerable effort to making smoking cessation permanent.

Although acutely ill patients have little appetite, tasty food in modest amounts is good for morale. Patients are usually offered a soft diet of 1500 to 1800 kcal/day with sodium reduction to 2 to 3 g. Sodium reduction is not required after the first 2 or 3 days if there is no evidence of heart failure. Patients are given a diet low in cholesterol and saturated fats, which is used to teach healthy eating.

For patients with diabetes and STEMI, intensive glucose control is not recommended; guidelines call for an insulin-based regimen to achieve and maintain glucose levels < 180 to 200 mg/dL (9.9 to 11.1 mmol/L) while avoiding hypoglycemia (9, 10, 11).

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Post-ACS Treatment and Rehabilitation

- Medications: Continuation of antiplatelets, beta-blockers, ACE inhibitors, and statins
- Sometimes functional evaluation
- Cardiac rehabilitation
- Changes in lifestyle: Regular exercise, diet modification, weight loss, smoking cessation

Medications

Several medications clearly reduce mortality risk post-MI and are used unless contraindicated or not tolerated:

- Antiplatelet therapy
- Beta-blockers, in patients with reduced ejection fraction
- Angiotensin-converting enzyme (ACE) inhibitors or angiotensin II receptor blockers (ARBs)
- Statins

Other medications that can reduce cardiovascular events include:

- PCSK9 inhibitors, ezetimibe, or bempedoic acid, which can be added to statins when further reduction in lipid levels is needed

Antiplatelet therapy is recommended to reduce mortality and reinfarction rates in patients after myocardial infarction (1). Dual antiplatelet therapy with aspirin and a P2Y12 receptor blocker (eg, ticagrelor, clopidogrel) for up to 1 year is recommended (1, 2, 3). A proton pump inhibitor should be given with dual antiplatelet therapy. If bleeding risk is a concern after PCI, patients can be transitioned to monotherapy with aspirin or a P2Y12 inhibitor after 1 month. Patients requiring long-term anticoagulation (eg, patients with atrial fibrillation) who are on "triple" therapy (dual antiplatelet plus an anticoagulant) after PCI can be transitioned to clopidogrel antiplatelet monotherapy along with their oral anticoagulant after 1 to 4 weeks. Having an option other than aspirin for long-term antiplatelet monotherapy represents a change from historical practice.

Beta-blockers are considered standard therapy in patients with reduced ejection fraction (LV ejection fraction ≤ 40%) but not in patients with preserved ejection fraction (1, 4, 5, 6). Most available beta-blockers (eg, acebutolol, atenolol, metoprolol, propranolol, timolol) reduce post-MI mortality rate by approximately 25% (7), but major trials have demonstrated that this benefit does not apply to those with preserved ejection fraction.

ACE inhibitors or ARBs are considered standard therapy for post-MI patients with LV ejection fraction < 40%, diabetes, hypertension, or an anterior STEMI. These medications may provide long-term cardioprotection by improving endothelial function; this benefit is still seen but may be less pronounced in the absence of heart failure, hypertension, diabetes, or chronic kidney disease (8).

Statins are also standard therapy and are routinely prescribed for MI patients with coronary artery disease, regardless of lipid levels (1, 9). Reducing cholesterol levels after MI reduces rates of recurrent ischemic events and mortality in patients with elevated or normal cholesterol levels. The statin should be continued indefinitely, unless significant adverse effects occur, and dose should be increased to the maximally tolerated dose. A non-statin lipid-lowering medication (such as ezetimibe, a PSCK9 inhibitor, or bempedoic acid) should be added for patients whose low-density lipoprotein (LDL) level remains ≥ 55 to 70 mg/dL (≥ 1.81 mmol/L), depending on the individual's target level, after 4 to 8 weeks on maximally tolerated statin therapy, or for those who are not able to tolerate a statin at all.

Mineralocorticoid receptor antagonists (spironolactone or eplerenone) are indicated for patients with left ventricular dysfunction and either heart failure symptoms or diabetes following ACS, to reduce morbidity and mortality (1, 10). In patients with heart failure, data also support sodium–glucose co-transporter 2 (SGLT2) inhibition after MI, regardless of diabetes status, to reduce the risk of worsening heart failure, cardiovascular mortality, or both (11).

Functional evaluation

Patients who did not have coronary angiography during admission, have no high-risk features (eg, heart failure, recurrent angina, ventricular tachycardia or ventricular fibrillation after 24 hours, mechanical complications such as new murmurs, shock), and have an ejection fraction > 40% usually should have stress testing of some sort before or shortly after discharge (see table Functional Evaluation After Acute Coronary Syndrome).

TABLE

Functional Evaluation After Acute Coronary Syndrome

Exercise Capacity	If ECG Is Interpretable	If ECG Is Not Interpretable
Able to exercise	Submaximal or symptom-limited stress ECG before or after discharge	Exercise echocardiography or nuclear scanning
Unable to exercise	Pharmacologic stress testing (echocardiography or nuclear scanning)	Pharmacologic stress testing (echocardiography or nuclear scanning)

Activity

Physical activity is gradually increased during the first 3 to 6 weeks after discharge. Resumption of sexual activity, often of importance to the patient and partner, and other moderate physical activities may be encouraged. If good cardiac function is maintained 6 weeks after acute myocardial infarction, most patients can return to all their normal activities. A regular exercise program consistent with lifestyle, age, and cardiac status reduces risk of ischemic events and enhances general well-being. Supervised cardiac rehabilitation programs based in clinics or hospitals or home-based programs are recommended for all patients after acute coronary syndromes (1); they decrease cardiovascular mortality after revascularization (12).

Risk factor modification

The acute illness and treatment of ACS should serve as a catalyst for discussion of modifiable cardiovascular risk factors. Evaluating the patient's physical and emotional statuses and discussing them with the patient, advising about lifestyle (eg, smoking, diet, work and play habits, exercise), and aggressively managing risk factors may improve prognosis.

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Key Points

- **Non-ST-segment elevation acute coronary syndromes (NSTE-ACS, including unstable angina, non–ST-segment elevation myocardial infarction [NSTEMI]), and ST-segment elevation myocardial infarction (STEMI) represent a continuum of myocardial ischemia and necrosis; the distinctions help differentiate prognosis and guide treatment.**
- **Diagnosis is based on serial ECG and cardiac biomarker levels, particularly high-sensitivity cardiac troponin (hs-cTn) assays.**
- **Immediate medical treatment depends on the specific syndrome and patient characteristics but typically involves antiplatelet agents, anticoagulants, and nitrates as needed (eg, for chest pain, hypertension, pulmonary edema).**
- **For STEMI, do emergency PCI when door to balloon-inflation time is < 90 minutes (120 minutes accounting for hospital to hospital transfers); do fibrinolysis if such timely PCI is not available.**

- **For intermediate to high-risk NSTE-ACS, do angiography within 24 to 72 hours of hospitalization to identify coronary lesions requiring PCI or coronary artery bypass grafting (CABG); fibrinolysis is not indicated.**
- **For low-risk NSTE-ACS (including unstable angina), use a selective invasive strategy with stress testing, coronary CT angiography, or angiography prior to discharge.**
- **Following recovery, initiate or continue dual antiplatelet therapy, beta-blockers, angiotensin-converting enzyme inhibitors or angiotensin II receptor blockers, and statins in most cases, unless contraindicated.**

Drug Information for the Topic

MERCK MANUAL
Professional Version

Acute Myocardial Infarction (MI)

*Full Review: Jul 2026 By **Ranya N. Sweis**, MD, MS, Northwestern University Feinberg School of Medicine | **Arif Jivan**, MD, PhD, Northwestern University Feinberg School of Medicine | Peer reviewed by **Jonathan G. Howlett**, MD, Cumming School of Medicine, University of Calgary*
Last updated: Jul 2026

Acute myocardial infarction is myocardial necrosis, usually resulting from acute obstruction of a coronary artery. Symptoms include chest discomfort with or without dyspnea, nausea, and/or diaphoresis. Diagnosis is by electrocardiography (ECG) and a significant elevation in cardiac troponin. Immediate treatment is with antiplatelets, anticoagulants, nitrates, and reperfusion therapy. For ST-segment-elevation myocardial infarction, emergency reperfusion is via percutaneous intervention, fibrinolytic drugs if percutaneous intervention is not available, or occasionally coronary artery bypass graft surgery. For non–ST-segment-elevation myocardial infarction, reperfusion is via percutaneous intervention or coronary artery bypass graft surgery. Subsequent care includes lipid management, beta-blockers, cardiac rehabilitation, risk factor management, and dual antiplatelet therapy.

In the United States, approximately 800,000 myocardial infarctions (MI) occur annually (1), resulting in death for approximately 100,000 people. Worldwide, 1.4 million deaths are attributed to ischemic heart disease, making it the leading cause of death globally (2).

Acute MI, along with unstable angina, is considered an acute coronary syndrome. Acute MI includes both non–ST-segment elevation myocardial infarction (NSTEMI) and ST-segment elevation myocardial infarction (STEMI). Distinction between NSTEMI and STEMI is vital as treatment strategies and timing of treatment are different for these 2 entities. NSTEMI is classified, along with unstable angina, as a non-ST-segment elevation acute coronary syndrome (NSTE-ACS) and shares the same diagnostic, risk stratification, and treatment pathway.

Heart Attack and Cardiac Arrest Myths

PODCAST

General references

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Pathophysiology of Acute MI

Myocardial infarction is defined as myocardial necrosis in a clinical setting consistent with myocardial ischemia (1). These conditions can be satisfied by a rise of cardiac troponin (cTn) above the 99th percentile of the upper reference limit (URL) plus at least one of the following:

- Symptoms of ischemia
- ECG changes indicative of new ischemia (significant ST-segment or T wave changes, or new left bundle branch block in the appropriate clinical context)
- Development of pathologic Q waves
- Imaging evidence of new loss of myocardium or new regional wall motion abnormality
- Angiography or autopsy evidence of intracoronary thrombus

Slightly different criteria are used to diagnose MI during and after percutaneous coronary intervention or coronary artery bypass grafting, and as the cause of sudden death.

MI can be classified based on etiology and circumstances (1):

- Type 1: Spontaneous MI caused by ischemia due to a primary coronary event (eg, plaque rupture, erosion, or fissuring; coronary dissection)
- Type 2: Ischemia due to increased oxygen demand (eg, hypertension) or decreased supply (eg, coronary artery spasm or embolism, arrhythmia, hypotension)
- Type 3: Related to sudden unexpected cardiac death

- Type 4a: Associated with percutaneous coronary intervention (signs and symptoms of myocardial infarction with cTn values > 5 × 99th percentile URL)
- Type 4b: Associated with documented stent thrombosis
- Type 5: Associated with coronary artery bypass grafting (signs and symptoms of myocardial infarction with cTn values > 10 × 99th percentile URL)

Infarct location

MI affects predominantly the left ventricle (LV), but damage may extend into the right ventricle (RV) or the atria.

Right ventricular infarction usually results from obstruction of the proximal right coronary artery or a marginal branch; it is characterized by high RV filling pressure, often with severe tricuspid regurgitation and reduced cardiac output.

An **inferoposterior infarction**, usually reflecting right coronary or dominant left circumflex artery obstruction, causes some degree of RV dysfunction in about half of patients, with a hemodynamic abnormality in a significant fraction of those (2, 3). RV dysfunction should be considered in any patient who has inferoposterior infarction and elevated central venous pressure with hypotension or shock. RV infarction complicating LV infarction is associated with increased mortality and risk of shock and arrhythmia (4).

Anterior infarcts tend to be larger and result in a worse prognosis than inferoposterior infarcts. They are usually due to left coronary artery obstruction, especially in the anterior descending artery.

Infarct extent

Infarction may be:

- Transmural
- Nontransmural

Transmural infarcts involve the whole thickness of myocardium from epicardium to endocardium and are usually characterized by ST-segment elevation and, later, by the development of abnormal Q waves on ECG.

Nontransmural (including subendocardial) infarcts do not extend through the ventricular wall and cause only ST-segment and T-wave (ST-T) abnormalities. Subendocardial infarcts usually involve the inner one-third of myocardium, where wall tension is highest and myocardial blood flow is most vulnerable to circulatory changes. These infarcts may follow prolonged hypotension.

Because the transmural depth of necrosis cannot be precisely determined clinically, infarcts are usually classified as STEMI or NSTEMI by the presence or absence of ST-segment elevation or Q waves on the ECG. Volume of myocardium destroyed can be roughly estimated by cardiac troponin levels and evaluated more precisely by later cardiac MRI.

Non–ST-segment elevation myocardial infarction (NSTEMI, subendocardial MI) is myocardial necrosis (evidenced by elevated cardiac troponin in blood) without acute ST-segment elevation. ECG changes such as ST-segment depression, T-wave inversion, or both may be present. NSTEMI is usually not transmural, although approximately 25% of NSTEMI are associated with a total occlusion of a coronary artery or branch and thus have the potential to become transmural if treatment is delayed (5).

ST-segment elevation myocardial infarction (STEMI, transmural MI) is myocardial necrosis with ECG changes showing ST-segment elevation that is not quickly reversed by nitroglycerin. Troponin is elevated. STEMI is typically transmural, although timely reperfusion, especially with an infarct territory that is small or has collateral vessels, may lead to a non-transmural area of myonecrosis (6).

MRI Findings of Subendocardial Myocardial Infarct

IMAGE

This cardiac MRI shows the right ventricle (red arrow) and left ventricle (green arrow) separated by the septal myocardium (blue arrow). There is late gadolinium subendocardial enhancement characteristic of a small myocardial infarct in the lateral wall of the left ventricle (white arrow).

Myocardial infarction in the absence of obstructive coronary artery disease (MINOCA)

Myocardial infarction in the absence of obstructive coronary artery disease (MINOCA) is found in approximately 5 to 6% of patients with acute MI who undergo coronary angiography (7). Patients with MINOCA tend to be younger, female, and without dyslipidemia. They tend to have myocardial necrosis without significant coronary atherosclerosis. Plaque disruption in arteries without significant stenosis, coronary vasospasm, coronary thrombosis or embolism, and spontaneous coronary artery dissection are causes of MINOCA. Medical management should be based on the underlying mechanism for MINOCA in each patient.

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Symptoms and Signs of Acute MI

Symptoms of NSTEMI and STEMI are the same. Days to weeks before the event, patients may experience prodromal symptoms, including unstable or crescendo angina, dyspnea, fatigue, sleep disturbances, anxiety, headache, and gastrointestinal complaints (1).

Often, the first symptom of infarction is deep, substernal, visceral pain, described as aching or pressure, often radiating to the back, jaw, left arm, right arm, shoulders, or all of these areas. The pain is similar to stable angina but is usually more severe and long-lasting; more often accompanied by dyspnea, diaphoresis, nausea, and/or vomiting; and relieved little or only temporarily by rest or nitroglycerin.

However, discomfort may be mild; in the general population, approximately 20% of acute MIs are silent (2, 3), and up to 30% may be silent in patients with diabetes. Silent MIs are those that are asymptomatic or are causing vague symptoms that are not recognized as illness by the patient. Silent MIs are also more common in patients with a history of coronary artery disease. Patients often interpret their discomfort as indigestion, particularly because spontaneous relief may be falsely attributed to belching or antacid consumption.

Silent ischemia (which can cause a silent MI) sometimes manifests as transient asymptomatic ST-T abnormalities seen during stress testing or 24-hour Holter monitoring. Radionuclide studies can sometimes document asymptomatic myocardial ischemia during physical or mental stress. Silent ischemia and stable angina may coexist, occurring at different times.

Some patients present with syncope.

Chest pain is the predominant symptom in both males and females, but females are more likely to experience additional symptoms and atypical chest pain, and they are less likely to have prodromal symptoms attributed to heart disease (4). Older patients may report dyspnea more than ischemic-type chest pain (5).

In severe ischemic episodes, the patient often has significant pain and feels restless and apprehensive. Nausea and vomiting may occur, especially with inferior MI. Dyspnea and weakness due to LV failure, pulmonary edema, shock, or significant arrhythmia may dominate.

Skin may be pale, cool, and diaphoretic. Peripheral or central cyanosis may be present. Pulse may be thready, and blood pressure is variable, although many patients initially have some degree of hypertension during pain.

Heart sounds are usually somewhat distant; a fourth heart sound (S4) is almost universally present. A soft systolic blowing apical murmur (reflecting papillary muscle dysfunction causing mitral regurgitation) may occur. During initial examination, a friction rub or more striking murmurs suggest a preexisting heart disorder or another diagnosis. Detection of a friction rub within a few hours after onset of MI symptoms suggests acute pericarditis rather than MI. However, friction rubs, usually evanescent, are common on days 2 and 3 post-STEMI. In some patients, the chest wall is tender when palpated.

In right ventricular (RV) infarction, signs include elevated RV filling pressure, distended jugular veins (often with Kussmaul sign), clear lung fields, and hypotension.

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Diagnosis of Acute MI

- Serial ECGs

- Serial cardiac troponin assays, preferably high-sensitivity
- Immediate coronary angiography (unless fibrinolytics are given) for patients with STEMI or NSTE-ACS with complications (eg, refractory chest pain, new or worsening heart failure, hemodynamic instability, cardiogenic shock, or unstable arrhythmias)
- Delayed coronary angiography for patients with NSTEMI (< 24 hours for high-risk, < 72 hours for intermediate risk)

Evaluation begins with initial and serial ECG and serial measurements of cardiac troponin to help distinguish between unstable angina, ST-segment elevation myocardial infarction (STEMI), and non–ST-segment elevation myocardial infarction (NSTEMI). Urgent or emergent cardiac catheterization is indicated for patients with acute STEMI but not generally for those with NSTEMI.

ECG

ECG is the most important test and should be performed as soon as possible (eg, within 10 minutes of presentation).

For **STEMI**, initial ECG is usually diagnostic, showing ST-segment elevation ≥ 1 mm in 2 or more contiguous leads subtending the damaged area; or in V2-V3 ≥ 2 mm in males 40 years or older, ≥ 2.5 mm in males younger than 40 years, and 1.5 mm in females of all ages (1). Reciprocal depression may be present in other leads. Pathologic Q waves often develop over time but are not necessary for the diagnosis (see figures Acute Lateral Left Ventricular Infarction, Lateral Left Ventricular Infarction, Lateral Left Ventricular Infarction (several days later), Acute Inferior (Diaphragmatic) Left Ventricular Infarction, Inferior (Diaphragmatic) Left Ventricular Infarction, and Inferior (Diaphragmatic) Left Ventricular Infarction (several days later)).

Acute Lateral Left Ventricular Infarction (Tracing Obtained Within a Few Hours of Illness Onset)

There is striking hyperacute ST-segment elevation in leads I, aVL, V_4, and V_6 and reciprocal depression in other leads.

Lateral Left Ventricular Infarction (After the First 24 Hours)

ST segments are less elevated; significant Q waves develop and R waves are lost in leads I, aVL, V_4, and V_6.

Lateral Left Ventricular Infarction (Several Days Later)

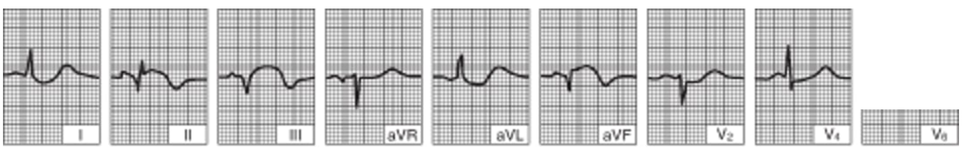
Significant Q waves and loss of R-wave voltage persist. ST segments are now essentially isoelectric. The ECG will probably change only slowly over the next several months.

Acute Inferior (Diaphragmatic) Left Ventricular Infarction (Tracing Obtained Within a Few Hours of Illness Onset)

There is hyperacute ST-segment elevation in leads II, III, and aVF and reciprocal depression in other leads.

Inferior (Diaphragmatic) Left Ventricular Infarction (After the First 24 Hours)

Significant Q waves develop with decreasing ST-segment elevation in leads II, III, and aVF.



Inferior (Diaphragmatic) Left Ventricular Infarction (Several Days Later)

ST segments are now isoelectric. Abnormal Q waves in leads II, III, and aVF indicate that myocardial scars persist.

The ECG must be read carefully because ST-segment elevation may be subtle, particularly in the inferior leads (II, III, aVF); sometimes the reader's attention is mistakenly focused on leads with ST-segment depression. If symptoms are present, ST-segment elevation on ECG has a specificity of 98 to 99% and a sensitivity of 35 to 56% for diagnosing myocardial infarction (2, 3). Serial tracings (obtained every 8 hours for 1 day, then daily) showing a gradual evolution toward a stable, more normal pattern or development of abnormal Q waves over a few days tends to confirm the diagnosis.

Left-sided posterior leads (V7-V9) should be used to detect left circumflex occlusion when ST-elevation in V1-V3 is present but does not meet criteria for STEMI (1). If right ventricular (RV) infarction is suspected ; additional leads are placed at V4-6R.

Right-Sided (V1R–V6R) and Posterior (V7–V9) Leads

IMAGE

The chest electrode of ECG includes conventional chest lead V6, posterior wall leads V7-V9 (right illustration), and right ventricular leads V1R-V6R (left illustration).

凌鲲 宋/STOCK.ADOBE.COM

For **NSTEMI**, characteristic ECG findings include new downsloping or horizontal ST depression (≥ 0.5 mm in at least 2 contiguous leads), T-wave inversion (> 1 mm in at least 2 contiguous leads with prominent R wave or R/S ratio > 1), or transient ST-segment elevation.

NSTEMI

IMAGE

This ECG of a patient in NSTEMI shows sinus tachycardia with right bundle branch block and inferolateral ST depressions and T wave inversions.

ECG diagnosis of MI is more difficult when a left bundle branch block configuration is present because it resembles STEMI changes. ST-segment elevation concordant with the QRS complex strongly suggests MI as does > 5-mm ST-segment elevation in at least 2 precordial leads. Although STEMI must be suspected in any patient with suggestive symptoms and new-onset (or not known to be old) left bundle branch block, a new left bundle branch block is not considered a STEMI equivalent in isolation (1, 4).

CLINICAL CALCULATORS

M.I. Criteria for Likelihood in Chest Pain with LBBB

Cardiac troponin

Cardiac troponins (cTn), preferably using a high-sensitivity assay (hs-cTN), are the biomarkers used in the diagnosis of myocardial infarction. An abnormal cTn level, using an appropriate cutoff, is the primary distinguishing factor between unstable angina and myocardial infarction, particularly when ST elevation is not present.

Patients suspected of having an acute coronary syndrome should have an hs-cTn assay performed on presentation and again 1 to 2 hours later. Troponin should be measured at 0 and 3 to 6 hours if a standard cTn assay is used. Because troponin elevation lags behind the onset of myocardial injury, angiography and treatment in patients with STEMI, or in unstable patients with a NSTE-ACS, should not be delayed because of an initially normal troponin level.

Standard cardiac troponin assays, which have been in use for many years, are sensitive and specific but unlikely to detect cardiac troponins except in patients who had an acute cardiac disorder such as a myocardial infarction. Thus, a "positive" cTn test (ie, above the limit of detection) is very specific. Highly sensitive assays of cardiac troponin (hs-cTn) can detect small amounts of troponin in many healthy people, and conditions besides a myocardial infarction can cause an elevation in troponin (see Causes of Elevated Troponin Levels). Thus, troponin levels detected with hs-cTn tests need to be referenced to the normal range, and findings are defined as "elevated" with regard to MI diagnosis only when higher than 99% of the reference population (1).

Although diagnostic criteria for MI are well-established (1), an hs-cTn assay must be interpreted based on the patient's pre-test probability of disease. A high pre-test probability plus an elevated troponin level detected with an hs-cTn assay is highly suggestive of myocardial infarction, whereas a low pre-test probability plus a normal hs-cTn assay result is unlikely to represent myocardial infarction. Diagnosis is more challenging when test results are discordant with pre-test probability, in which case serial hs-cTn assays often help. A patient with low pre-test probability and an initially slightly elevated troponin level detected with a hs-cTn assay that remains stable on repeat testing probably has non-ACS cardiac disease (eg, heart failure, stable coronary artery disease). However, if the repeat level rises significantly (ie, > 20 to 50%), the likelihood of myocardial infarction becomes much higher. If a patient with high pre-test probability has a normal troponin level detected with a hs-cTn assay that rises > 50% on repeat testing, myocardial infarction is likely; continued normal levels (often including at 6 hours and beyond when suspicion is high) suggest the need to pursue an alternate diagnosis.

Coronary angiography

Coronary angiography most often combines diagnosis with percutaneous coronary intervention (PCI—ie, angioplasty, stent placement). When possible, emergency coronary angiography and PCI are performed as soon as possible after the onset of acute myocardial infarction (primary PCI). In many studies, a shorter interval to PCI ("door to balloon" or "onset to balloon" time) is associated with significantly lower morbidity and mortality and improved long-term outcomes (5, 6, 7, 8).

Angiography is obtained urgently for patients with STEMI, and for those with cardiogenic shock, new or worsening heart failure, chest pain refractory to treatment, or hemodynamic instability (including due to arrhythmia). Patients with high- or intermediate-risk NSTEMI typically undergo angiography within the first 24 to 72 hours of hospitalization to detect lesions that may require treatment; those with low-risk NSTE-ACS, including unstable angina, may undergo angiography prior to discharge or noninvasive testing such as stress testing or coronary CT angiography, followed by angiography if indicated after noninvasive testing (eg by inducible myocardial ischemia on stress testing) (1).

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Treatment of Acute MI

- Prehospital care: Oxygen, aspirin, and nitrates and triage to an appropriate medical center
- In-hospital pharmacologic therapy: Additional antiplatelet agents, analgesics, anticoagulants, and in some cases other medications
- Usually, reperfusion with percutaneous coronary intervention (PCI), coronary artery bypass grafting (CABG), or fibrinolytics for STEMI when PCI is not available
- Supportive care
- Post-discharge cardiac rehabilitation and chronic management of coronary artery disease

Choice of pharmacologic therapy and choice of reperfusion strategy are discussed elsewhere.

Prehospital care

- Oxygen
- Aspirin
- Nitrates
- Triage to appropriate medical center

A reliable IV route must be established, oxygen given (typically 2 L by nasal cannula), and continuous ECG monitoring started. Prehospital interventions by emergency medical personnel (including ECG, chewed aspirin [160 to 325 mg], and pain management with nitroglycerin) can reduce risk of mortality and complications (1). Early diagnostic data and response to treatment can help determine the need for and timing of revascularization, and guide triage to the appropriate hospital.

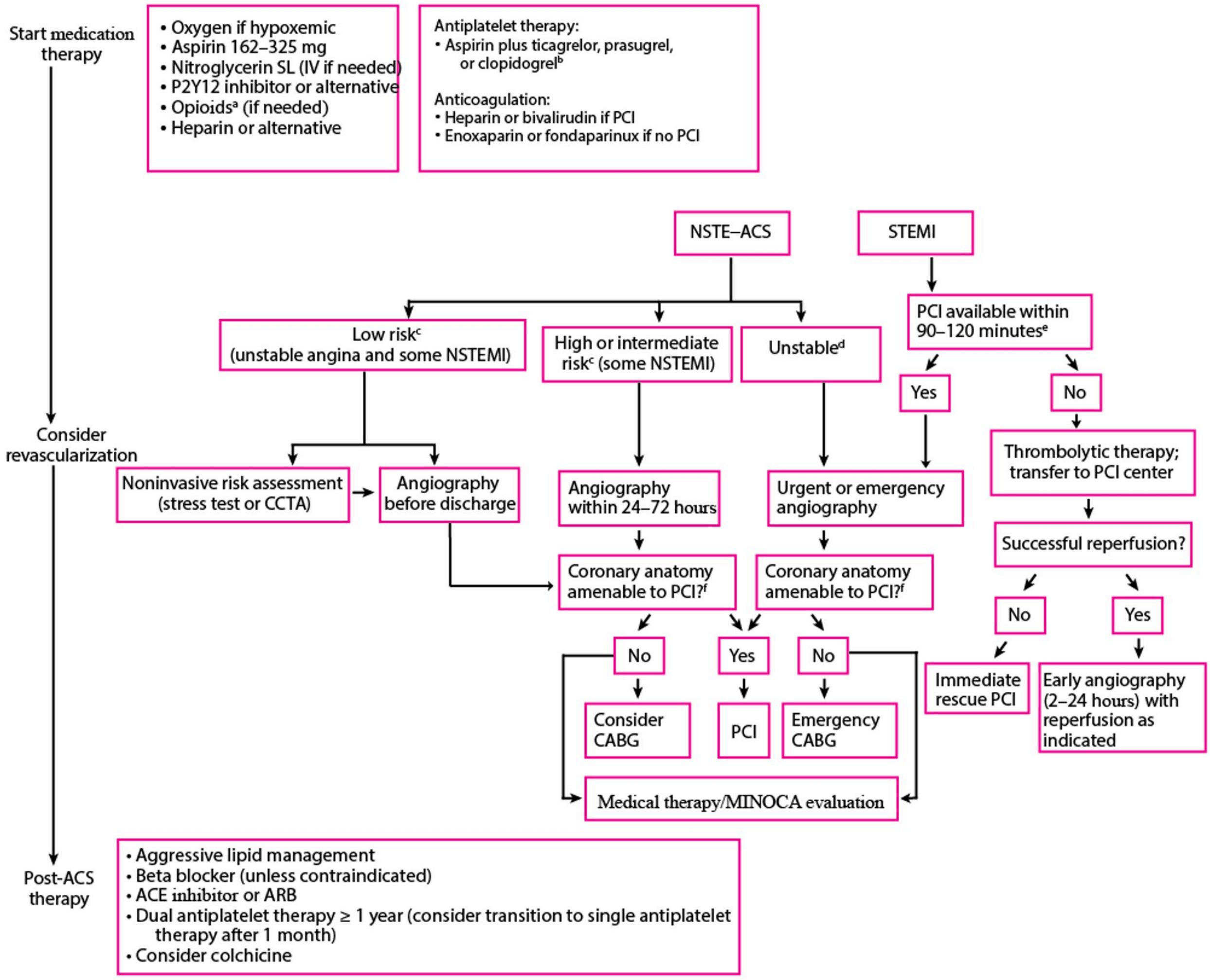
Hospital admission

- Risk-stratify patient and choose reperfusion strategy (PCI, CABG, or fibrinolytic therapy)
- Pharmacologic therapy with antiplatelets, anticoagulants and other medications based on reperfusion strategy

On arrival to the emergency department, the patient's diagnosis is confirmed. Pharmacologic therapy and timing of revascularization depend on the clinical picture and diagnosis.

For STEMI, reperfusion strategy can include immediate PCI, or fibrinolytic therapy when PCI is not available within 90 to 120 minutes. For patients with NSTEMI classified as intermediate- or high-risk (see Risk Stratification in ACS), angiography may be performed within 24 to 72 hours of admission if the patient is clinically stable. If the patient is unstable (eg, ongoing symptoms, hypotension, or sustained arrhythmias), angiography must be done immediately (see figure Approach to Acute Myocardial Infarction).

Approach to Acute Myocardial Infarction



a Use morphine or fentanyl judiciously if nitroglycerin is contraindicated or if the patient has symptoms despite nitroglycerin therapy; they may attenuate the effect of P2Y12 inhibitors.

b See Antiplatelet Agents for more detail.

c Based on risk scores, troponin, symptoms, and ECG changes.

d Unstable patients include those with cardiogenic shock, new or worsening heart failure symptoms, refractory ischemic symptoms, and hemodynamic or electrical instability.

e 90 minutes if patient is in the United States and calls 911 or presents to a PCI-capable hospital; 120 minutes with hospital-to-hospital transfer.

f CABG may be preferred to PCI for patients with the following: High complexity coronary disease that involves the left main coronary artery, diabetes and multivessel disease involving

the left anterior descending artery, multivessel or diffuse disease, or severe left ventricular dysfunction with multivessel or complex left main disease.

ACE = angiotensin-converting enzyme inhibitor; ARB = angiotensin II receptor blocker; CABG = coronary artery bypass grafting; CCTA = coronary CT angiography; NSTE–ACS = non–ST-segment elevation–acute coronary syndrome; MINOCA = myocardial infarction with non-obstructive coronary arteries; PCI = percutaneous intervention; SL = sublingual; STEMI = ST-segment elevation myocardial infarction.

Data from Rao SV, O'Donoghue ML, Ruel M, et al. 2025 ACC/AHA/ACEP/NAEMSP/SCAI Guideline for the Management of Patients With Acute Coronary Syndromes: A Report of the American College of Cardiology/American Heart Association Joint Committee on Clinical Practice Guidelines. *Circulation.* 2025;151(13):e771-e862. doi:10.1161/CIR.0000000000001309.

Pharmacologic treatment of acute myocardial infarction

All patients should be given antiplatelet agents, anticoagulants, and if chest pain is present, antianginal medications. The specific medications used depend on the reperfusion strategy and other factors; medication selection and use are discussed in Medications for Acute Coronary Syndromes. Other medications, such as beta-blockers, angiotensin-converting enzyme (ACE) inhibitors or angiotensin II receptor blockers (ARBs), and statins, should also be given (see Post-ACS Treatment of Acute Coronary Syndromes and the table Medications for Acute Coronary Syndromes and Chronic Coronary Artery Disease).

Patients with acute myocardial infarction should be given the following (unless contraindicated):

- Antiplatelet agents: Aspirin and prasugrel, ticagrelor, or clopidogrel
- Anticoagulants: A heparin (unfractionated or low molecular weight heparin) or bivalirudin
- Sometimes a glycoprotein IIb/IIIa inhibitor when PCI is performed
- Antianginal therapy, usually nitroglycerin
- Beta-blocker
- Angiotensin-converting enzyme (ACE) inhibitor or angiotensin II receptor blocker (ARB)
- Statin and sometimes additional lipid-lowering agents
- Sometimes mineralocorticoid receptor-antagonists or sodium-glucose co-transorter-2 (SGLT2) inhibitors

All patients are given aspirin 160 to 325 mg (not enteric-coated), if not contraindicated. Chewing the first dose before swallowing quickens absorption. Aspirin reduces short- and long-term mortality risk (2). In patients undergoing PCI, a loading dose of prasugrel (60 mg orally once), ticagrelor (180 mg orally once), or clopidogrel (300 to 600 mg orally once) is recommended to reduce ischemic events and improve outcomes. (3). For urgent PCI, prasugrel and ticagrelor are more rapid in onset and may be preferred. IV cangrelor may also be used during PCI to reduce ischemic events in patients who are P2Y12-inhibitor naive (4).

Either a low molecular weight heparin (LMWH), unfractionated heparin, or bivalirudin is given routinely to patients with myocardial infarction unless contraindicated (eg, by active bleeding) (4). Unfractionated heparin is more complicated to use because it requires frequent (every 6 hours) dosing adjustments to achieve target activated partial thromboplastin time (aPTT). The LMWHs have better bioavailability, are given by simple weight-based dose without monitoring aPTT and dose titration, and have lower risk of heparin-induced thrombocytopenia; they are recommended when an early invasive strategy is not planned. Bivalirudin is recommended for patients with a known or suspected history of heparin-induced thrombocytopenia who are undergoing PCI. Anticoagulants are continued for:

- Duration of PCI in patients undergoing this procedure
- Duration of hospital stay (in patients on LMWH) or 48 hours (in patients on unfractionated heparin) in all other cases

Consider a glycoprotein IIb/IIIa inhibitor during (or sometimes before) PCI for patients with high-risk lesions (high thrombus burden, no reflow). Abciximab, tirofiban, and eptifibatide appear to have equivalent efficacy, and the choice of medication should depend on other factors (eg, cost, availability, familiarity). Glycoprotein IIb/IIIa inhibitors are continued for 6 to 24 hours (4).

Chest pain can be treated with sublingual or intravenous nitroglycerin, or sometimes morphine or fentanyl. Nitroglycerin is preferable to opioids, which should be used judiciously (eg, if a patient has a contraindication to nitroglycerin or is in pain despite maximal nitroglycerin therapy) (4). Nitroglycerin is initially given sublingually, followed by continuous IV drip if needed. Morphine, given 2 to 4 mg IV, repeated every 15 minutes as needed, is highly effective but can depress respiration, can reduce myocardial contractility, and is a potent venous vasodilator. Evidence also suggests that morphine and fentanyl interfere with some P2Y12 receptor inhibitor activity. A large retrospective trial showed that morphine may increase mortality in patients with acute myocardial infarction (5, 6, 7). Hypotension and bradycardia secondary to morphine can usually be overcome by prompt elevation of the lower extremities.

Post-acute pharmacologic treatment

Dual antiplatelet therapy with aspirin and a P2Y12 inhibitor (eg, ticagrelor, clopidogrel) for up to 1 year is recommended (4, 8, 9). A proton pump inhibitor should be given with dual antiplatelet therapy. If PCI was performed, options for tapering to monotherapy sooner than 12 months, with aspirin or a P2Y12 inhibitor, are discussed in more detail under Treatment of Acute Coronary Syndromes and Medications for Acute Coronary Syndromes.

Standard therapy for all patients with acute coronary syndromes, including myocardial infarction, includes beta-blockers, ACE inhibitors or ARBs, and statins (3, 4). Beta-blockers are recommended unless contraindicated (eg, by bradycardia, heart block, hypotension, or asthma), especially for high-risk patients. Beta-blockers reduce heart rate, arterial pressure, and contractility, thereby reducing cardiac workload and oxygen demand. However, beta-blockers may have limited benefit in patients with preserved LV ejection fraction (10, 11). ACE inhibitors or ARBs may provide long-term cardioprotection by improving endothelial function (12), particularly in patients with heart failure, hypertension, diabetes, or chronic kidney disease. Statins are also standard therapy regardless of lipid levels (13) and should be continued indefinitely, with the addition of other lipid-lowering medications as needed.

Mineralocorticoid receptor antagonists (spironolactone or eplerenone) are indicated for patients with left ventricular dysfunction and either heart failure symptoms or diabetes following ACS, to reduce morbidity and mortality (4). In patients with heart failure, data also support blockade of the sodium–glucose co-transporter 2 (SGLT2) after MI, regardless of diabetes status, to reduce the risk of worsening heart failure, cardiovascular mortality, or both (4, 14).

Reperfusion therapy in acute myocardial infarction

- For patients with STEMI: Immediate percutaneous coronary intervention (PCI) or fibrinolytics if immediate angiography and PCI are not available at presentation hospital
- For patients with NSTEMI: Immediate PCI for unstable patients or within 24 to 48 hours for stable patients

For **patients with STEMI,** emergency PCI is the preferred treatment of ST-segment elevation myocardial infarction when available in a timely fashion (door to balloon-inflation time < 90 minutes, or < 120 minutes if hospital-to-hospital transfer is involved) by an experienced operator (4). If there is likely to be a significant delay in availability of PCI, thrombolysis should be performed for patients with STEMI meeting criteria (see Infarct extent). Reperfusion using fibrinolytics is most effective if given in the first few minutes to hours after onset of myocardial infarction. The earlier a fibrinolytic is begun, the better. The goal is a door-to-needle time of 30 to 60 minutes. Greatest benefit occurs within 3 hours, but the medications may be effective up to 12 hours. Characteristics and selection of fibrinolytic medications are discussed elsewhere. Patients with STEMI treated with fibrinolysis should undergo subsequent angiography with the intent to perform PCI between 2 and 24 hours afterwards in most patients but immediately if reperfusion with fibrinolytics failed.

Patients with unstable NSTE-ACS (including NSTEMI and unstable angina) (ie, those with shock, ongoing symptoms, hypotension, or sustained arrhythmias) should be taken directly to the cardiac catheterization laboratory for angiography to identify coronary lesions requiring PCI or coronary artery bypass grafting (CABG) (4).

For **patients with uncomplicated NSTEMI,** reperfusion is not as urgent because a completely occluded infarct-related artery at presentation is uncommon. Such patients typically undergo angiography within the first 24 to 72 hours of hospitalization to identify coronary lesions requiring PCI or CABG (4). Patients with low-risk NSTEMI may also undergo non-invasive risk stratification with stress testing or coronary CT angiography prior to or in lieu of angiography.

Fibrinolytics are not indicated for any patients with NSTE-ACS, including NSTEMI. Risk outweighs potential benefit.

Choice of reperfusion strategy is further discussed in Revascularization for Acute Coronary Syndromes.

Rehabilitation and post-discharge treatment

- Functional evaluation
- Cardiac rehabilitation
- Changes in lifestyle: Regular exercise, diet modification, weight loss, smoking cessation

Post-acute pharmacologic therapy is discussed elsewhere.

Patients who did not have coronary angiography during admission, have no high-risk features (eg, heart failure, recurrent angina, ventricular tachycardia or ventricular fibrillation after 24 hours, mechanical complications such as new murmurs, shock), and have an ejection fraction > 40% whether or not they received fibrinolytics usually should have stress testing of some sort before or shortly after discharge (see table Functional Evaluation After Myocardial Infarction).

TABLE

Functional Evaluation After Myocardial Infarction

Exercise Capacity	If ECG Is Interpretable	If ECG Is Not Interpretable
Able to exercise	Submaximal or symptom-limited stress ECG before or after discharge	Exercise echocardiography or nuclear scanning
Unable to exercise	Pharmacologic stress testing (echocardiography or nuclear scanning)	Pharmacologic stress testing (echocardiography or nuclear scanning)

Supervised cardiac rehabilitation programs based in clinics or hospitals or home-based programs are recommended for all patients after acute coronary syndromes (4); these programs decrease cardiovascular mortality after revascularization (15).

The acute illness and treatment of myocardial infarction should serve as a catalyst for discussion of modifiable cardiovascular risk factors. Evaluating the patient's physical and emotional status and discussing them with the patient, advising about lifestyle (eg, smoking, diet, work and play habits, exercise), and aggressively managing risk factors may improve prognosis.

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Prognosis for Acute MI

Risk should be estimated via formal clinical risk scores (eg, Thrombosis in Myocardial Infarction [TIMI]—see tables Mortality Risk at 30 Days in STEMI and Risk of Adverse Events at 14 Days in Unstable Angina or NSTEMI), the Global Registry of Acute Coronary Events [GRACE] score, or a combination of the following high-risk features:

- Recurrent angina/ischemia at rest or during low-level activity
- Heart failure
- Worsening mitral regurgitation
- High-risk stress test result (test stopped in ≤ 5 minutes due to symptoms, marked ECG abnormalities, hypotension, or complex ventricular arrhythmias)
- Hemodynamic instability
- Sustained ventricular tachycardia
- Diabetes mellitus
- PCI within past 6 months
- Prior coronary artery bypass grafting (CABG)
- LV ejection fraction < 0.40

The Killip classification categorizes patients with an acute MI based on physical examination findings suggestive of left ventricular failure, and higher scores predicting higher mortality risk (1) (see table Killip Classification and Mortality Rate of Acute MI), and is incorporated into both the TIMI and GRACE scores.

In-hospital mortality of myocardial infarction is approximately 3 to 8% (2, 3), compared with 30% prior to the widespread use of PCI and fibrinolytics. In-hospital mortality is generally lower for NSTEMI than for STEMI. Mortality rates tend to be higher in females and in patients with diabetes (4, 5).

Cardiogenic shock is the largest cause of in-hospital mortality in patients with STEMI undergoing PCI. Most fatalities among patients who do survive initial hospitalization occur in the first year after diagnosis (6). Factors associated with higher risk of adverse events in the year after discharge include decreased LV ejection fraction, progression of heart failure, diabetes, and poor renal function (7, 8). Assessment of ejection fraction, usually with echocardiography, is recommended prior to discharge after acute MI (9).

Many authorities recommend stress ECG before hospital discharge or within 6 weeks of MI. Good exercise performance without ECG abnormalities is associated with a favorable prognosis; further evaluation is usually not required. Poor exercise performance is associated with a poor prognosis.

Cardiac performance after recovery depends largely on how much functioning myocardium survives the acute attack. Acute damage adds to scars from previous infarcts.

TABLE

Killip Classification of Acute Myocardial Infarction*

Class	PAO_2†	Clinical Description

Class	PAO_2†	Clinical Description
1	Normal	No clinical evidence of left ventricular (LV) failure
2	Slightly reduced	Mild to moderate LV failure
3	Abnormal	Severe LV failure, pulmonary edema
4	Severely abnormal	Cardiogenic shock: hypotension, tachycardia, mental obtundation, cool extremities, oliguria, hypoxia

* Determined by repeated examination of the patient during the course of illness.

† Determined while the patient is breathing room air.

Modified from Killip T 3rd, Kimball JT. Treatment of myocardial infarction in a coronary care unit. A two-year experience with 250 patients. *Am J Cardiol.* 1967;20(4):457-464. doi:10.1016/0002-9149(67)90023-9

TABLE

Mortality Risk at 30 Days in STEMI

Risk Factor	Points
Age ≥ 75 years	3
Age 65–74 years	2
Diabetes mellitus, hypertension, or angina	1
Systolic blood pressure < 100 mm Hg	3
Heart rate > 100 beats/minute	2
Killip class II–IV	2
Weight < 67 kg	1
Anterior ST-elevation or left branch bundle block	1

Risk Factor	Points
Time to treatment > 4 hours	1
Total points possible	0–14
Risk	
Total Points	Mortality Rate at 30 Days (%)
0	0.8
1	1.6
2	2.2
3	4.4
4	7.3
5	12.4
6	16.1
7	23.4
8	26.8
> 8	35.9

MI = myocardial infarction; STEMI = ST-segment elevation MI; TIMI = thrombolysis in MI.

Based on Morrow DA, Antman EM, Charlesworth A, et al. TIMI risk score for ST-elevation myocardial infarction: a convenient, bedside, clinical score for risk assessment at presentation: An intravenous nPA for treatment of infarcting myocardium early II trial substudy. *Circulation.* 2000;102 (17):2031-2037. doi:10.1161/01.cir.102.17.2031 and Rao SV, O'Donoghue ML, Ruel M, et al. 2025 ACC/AHA/ACEP/NAEMSP/SCAI Guideline for the Management of Patients With Acute Coronary Syndromes: A Report of the American College of Cardiology/American Heart Association Joint Committee on Clinical Practice Guidelines. *Circulation.* 2025;151(13):e771-e862. doi:10.1161/CIR.0000000000001309.

TABLE

Risk of Adverse Events* at 14 Days in Unstable Angina or NSTEMI

Scoring

Risk Factor	Points
Age > 65 years	1

Scoring

Risk Factor	Points
CAD risk factors (must have ≥ 3 for 1 point): • Family history • Hypertension • Current smoking • High cholesterol • Diabetes mellitus	1
Known CAD (stenosis ≥ 50%)	1
Previous chronic use of aspirin	1
Two episodes of rest angina in past 24 hours	1
Elevated cardiac biomarkers	1
ST elevation ≥ 0.5 mm	1
Risk level is based on total points:	1–2 = low 3–4 = intermediate 5–7 = high
Absolute risk	
Total Points	Risk of Events at 14 Days (%)*
0 or 1	4.7
2	8.3
3	13.2
4	19.9
5	26.2
6 or 7	40.9

* Events include all-cause mortality, myocardial infarction, and recurrent ischemia requiring urgent revascularization.

CAD = coronary artery disease; MI = myocardial infarction; NSTEMI = non–ST-segment elevation MI; TIMI = thrombolysis in MI.

Based on Antman EM, Cohen M, Bernink PJ, et al. The TIMI risk score for unstable angina/non-ST elevation MI: A method of prognostication and therapeutic decision making. *JAMA.* 2000;284(7):835-842. doi:10.1001/jama.284.7.835

CLINICAL CALCULATORS

Thrombolysis in Myocardial Infarction (TIMI) Score for Unstable Angina Non ST Elevation Myocardial Infarction

CLINICAL CALCULATORS

Thrombolysis in Myocardial Infarction (TIMI) Score for ST Elevation Acute Myocardial Infarction

CLINICAL CALCULATORS

GRACE 2.0 Score for Acute Coronary Syndrome Prognosis (non-linear model 2014)

Prognosis references

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Key Points

- **Acute myocardial infarction (MI) is myocardial necrosis of heart muscle tissue, usually due to acute obstruction of a coronary artery.**
- **Infarction location (right ventricle, inferoposterior, or anterior) and extent (transmural or nontransmural) affect symptoms, treatment, and prognosis.**
- **Symptoms typically include substernal aching or pressure (with or without radiation to the neck, back, or arms) dyspnea, diaphoresis, nausea, and/or vomiting.**
- **Females, older adults, and patients with diabetes are more likely to have atypical symptoms, and 20% of acute MIs are silent (asymptomatic).**
- **Diagnose with serial ECGs and cardiac troponin.**
- **Treat immediately with oxygen, antiplatelets, nitrates, and anticoagulants.**
- **For patients with ST-segment elevation MI (STEMI), treat with immediate angiography and percutaneous coronary intervention (PCI); if immediate PCI is not available, give intravenous fibrinolytics.**
- **For most patients with non-ST-segment elevation MI (NSTEMI), perform angiography and PCI within 24 to 72 hours if stable; do immediate PCI for unstable patients (persistent chest pain, shock, hypotension, unstable arrhythmias, or markedly elevated biomarkers).**
- **Initiate or continue antiplatelets, beta-blockers, ACE inhibitors (or angiotensin II receptor blockers), and statins following initial treatment.**

- **After hospital discharge, arrange functional evaluation and cardiac rehabilitation, and manage risk factors (eg, hypertension, diabetes, smoking), and promote healthy diet and exercise.**

Drug Information for the Topic

Medications for Acute Coronary Syndromes

*Full Review: Jul 2026 By **Ranya N. Sweis**, MD, MS, Northwestern University Feinberg School of Medicine | **Arif Jivan**, MD, PhD, Northwestern University Feinberg School of Medicine | Peer reviewed by **Jonathan G. Howlett**, MD, Cumming School of Medicine, University of Calgary*
Last updated: Jul 2026

Treatment of acute coronary syndromes (ACS) is designed to relieve distress, interrupt thrombosis, reverse ischemia, limit infarct size, reduce cardiac workload, and prevent and treat complications. An ACS is a medical emergency; outcome is greatly influenced by rapid diagnosis and treatment. Treatment occurs simultaneously with diagnosis. Treatment includes revascularization (with percutaneous coronary intervention [PCI], coronary artery bypass grafting [CABG], or fibrinolytic therapy) and pharmacologic therapy to treat ACS and underlying coronary artery disease.

Medications used depend on the type of ACS and include:

- Antiplatelet therapy: Aspirin and P2Y12 inhibitors (clopidogrel, prasugrel, ticagrelor, cangrelor)
- Nitroglycerin (sublingual or IV) and sometimes opioids
- Anticoagulation (unfractionated or low molecular weight heparin) or bivalirudin (particularly in patients with ST-segment elevation myocardial infarction [STEMI] who are at high risk of bleeding)
- Glycoprotein IIb/IIIa inhibitor for certain patients undergoing PCI and high-risk lesions (eg, high thrombus burden, no reflow)
- Fibrinolytics for select patients with ST-segment elevation myocardial infarction (STEMI) when timely PCI unavailable
- Beta-blocker
- Statins and sometimes other lipid-lowering agents
- Angiotensin-converting enzyme (ACE) inhibitor or angiotensin II receptor blocker (ARB)
- Sometimes sodium-glucose-cotransporter 2 (SGLT2) inhibitor
- Sometimes mineralocorticoid receptor antagonist (MRA)

Antiplatelet and antithrombotic medications, which stop clots from forming, are used routinely. Anti-ischemic medications (eg, beta-blockers, IV nitroglycerin) are frequently added, particularly when chest pain or hypertension is present (see table Medications for Acute Coronary Syndromes and Chronic Coronary Artery Disease).

If primary PCI is not immediately available, fibrinolytics should be used if not contraindicated for STEMI, but fibrinolytics worsen outcome for unstable angina and non-ST elevation myocardial infarction

(NSTEMI) (1).

Chest pain can be treated with nitroglycerin or sometimes morphine or fentanyl. Nitroglycerin is preferable to morphine or fentanyl, which should be used judiciously (eg, if a patient has a contraindication to nitroglycerin or is in pain despite nitroglycerin therapy). Nitroglycerin is initially given sublingually, followed by continuous IV drip if needed. Morphine is highly effective but can depress respiration, can reduce myocardial contractility, and is a potent venous vasodilator. Evidence also suggests that morphine and fentanyl interfere with some P2Y12 receptor inhibitors (2, 3, 4). A large retrospective trial showed that morphine may increase mortality in patients with acute myocardial infarction (3, 4). Hypotension and bradycardia secondary to opioids can usually be overcome by prompt elevation of the lower extremities.

Blood pressure (BP) is normal or slightly elevated in most patients on arrival at the emergency department; BP gradually falls over the next several hours. Continued hypertension requires treatment with antihypertensives, preferably IV nitroglycerin, to lower BP and reduce cardiac workload. Severe hypotension or other signs of shock are ominous and must be treated aggressively with IV fluids and sometimes vasopressors.

Post-acute management typically includes beta blockers and aggressive lipid management with statins and other agents. Longer-term therapy includes continuation of antiplatelet therapy, renin-angiotensin system inhibition with an ACE inhibitor or ARB, and sometimes a MRA, SGLT2 inhibitor, or colchicine.

See also Treatment of Stable Angina, Treatment of Acute Coronary Syndromes, Treatment of Unstable Angina, and Treatment of Acute Myocardial Infarction for more details on the clinical pharmacologic approach to each condition; and table Medications for Acute Coronary Syndromes and Chronic Coronary Disease and the text below for more information about specific medications.

TABLE

Medications for Coronary Artery Disease*

Medication	Dosage	Use
Angiotensin-converting enzyme (ACE) inhibitors		
Benazepril Captopril Enalapril Fosinopril Lisinopril Moexipril Perindopril Quinapril	Variable†	All patients with CAD, especially those with large infarctions, renal insufficiency, heart failure, hypertension, or diabetes Contraindications include hypotension, hyperkalemia, bilateral renal artery stenosis, pregnancy, and known allergy

Medication	Dosage	Use
Ramipril Trandolapril		
Angiotensin II receptor blockers (ARBs)		
Candesartan Eprosartan Irbesartan Losartan Olmesartan Telmisartan Valsartan	Variable†	An effective alternative for patients who cannot tolerate ACE inhibitors (eg, because of cough); currently, not first-line treatment after MI Contraindications include hypotension, hyperkalemia, bilateral renal artery stenosis, pregnancy, and known allergy
Anticoagulants: Direct-acting oral anticoagulants‡		
Apixaban	5 mg orally twice a day	May be useful long-term in patients with non-valvular atrial fibrillation
Dabigatran	150 mg orally twice a day (or 110–150 mg orally twice a day for patients also taking P2Y12 inhibitors)	
Edoxaban	60 mg orally once a day	
Rivaroxaban	20 mg orally once a day (or 15 mg orally once a day for patients also taking P2Y12 inhibitors)	
Anticoagulants: Direct thrombin inhibitors		
Argatroban	350 mcg/kg (IV bolus) followed by 25 mcg/kg/minute (IV infusion)	As an alternative to heparin in patients with ACS and a known or suspected history of heparin-induced thrombocytopenia
Bivalirudin	Variable†	
Anticoagulants: Factor Xa inhibitor		

Medication	Dosage	Use
Fondaparinux	2.5 mg subcutaneously every 24 hours	As an alternative to heparin in patients with ACS and a known or suspected history of heparin-induced thrombocytopenia
Anticoagulants: Low molecular weight heparins		
Dalteparin Enoxaparin¶ Tinzaparin	Variable†	Patients with unstable angina or NSTEMI Patients < 75 years receiving tenecteplase Almost all patients with STEMI as an alternative to unfractionated heparin (unless PCI is indicated and can be done in < 90 minutes); drug continued until PCI or CABG is done or patient is discharged
Anticoagulants: Unfractionated heparin		
Unfractionated heparin	**Patients with unstable angina or NSTEMI**: 60–70 units/kg IV (maximum, 5000 units; bolus), followed by 12–15 units/kg/hour (maximum, 1000 units/hour) for 48 hours or until PCI is complete **Patients with STEMI**: 60 units/kg IV (maximum, 4000 units; bolus) given when alteplase, reteplase, or tenecteplase is started, then followed by 12 units/kg/hour (maximum, 1000 units/hour) for 48	Patients with unstable angina or NSTEMI as an alternative to enoxaparin Patients who have STEMI and undergo urgent angiography and PCI or patients > 75 years receiving tenecteplase

Medication	Dosage	Use
	hours or until PCI is complete	
Anticoagulants: Vitamin K inhibitor		
Warfarin	Oral dose adjusted to maintain INR of 2–3	Recommended for primary prevention in patients at high risk of systemic emboli (ie, with atrial fibrillation, mechanical heart valves, venous thromboembolism, hypercoagulable disorders, or LV thrombus) May be useful for primary prevention in patients with STEMI and anterior wall akinesis or dyskinesis if risk of bleeding is low Reasonable for patients with asymptomatic mural thrombus
Antiplatelet agents		
Aspirin	**For stable angina†:** 75 or 81 mg orally once a day (enteric-coated) **For ACS:** 160–325 mg orally chewed (not enteric-coated) on arrival at emergency department and once a day thereafter during hospitalization and 81 mg§ orally once a day long-term after discharge	All patients with CAD unless aspirin is not tolerated or is contraindicated; used long-term
Clopidogrel	75 mg orally once a day **For patients undergoing PCI:** 300–600 mg orally once, then 75 mg orally	Used with aspirin or, in patients who cannot tolerate aspirin, alone

Medication	Dosage	Use
	once a day for 1–12 months	For elective PCI, maintenance therapy required for at least 1 month for bare-metal stents and for at least 6–12 months for drug-eluting stents For ACS, dual antiplatelet therapy (typically with aspirin) is recommended for at least 12 months (for any type of stent)
Prasugrel	60 mg orally once before PCI, followed by 10 mg orally once a day for 1–12 months	Only for patients with ACS undergoing PCI Not used in combination with fibrinolytic therapy
Ticagrelor	**For patients undergoing PCI:** 180 mg orally once before the procedure, followed by 90 mg orally twice a day for 1–12 months	—
Ticlopidine	250 mg orally twice a day for 1–12 months	Rarely used routinely because neutropenia is a risk and white blood cell count must be monitored regularly
Beta-blockers		
Atenolol	Variable†	All patients with ACS, unless a beta-blocker is not tolerated or is contraindicated, especially high-risk patients; used long-term Intravenous beta-blockers may be used in patients with ongoing chest pain despite usual measures, or
Bisoprolol	Variable†	

Medication	Dosage	Use
Carvedilol	Variable†	persistent tachycardia, or hypertension in patients with unstable angina and myocardial infarction. Caution is necessary in patients with hypotension or other evidence of hemodynamic instability.
Metoprolol	Variable†	
Calcium channel blockers		
Amlodipine	Variable†	Patients with stable angina if symptoms persist despite nitrates use or if nitrates are not tolerated
Diltiazem (extended-release)	Variable†	
Felodipine	Variable†	
Nifedipine (extended-release)	Variable†	
Verapamil (extended-release)	Variable†	
Glycoprotein IIb/IIIa inhibitors		
Abciximab	Variable†	Some patients with ACS, particularly those who are having PCI with stent placement and high-risk patients with unstable angina or NSTEMI and large thrombus burden
Eptifibatide	Variable†	
Tirofiban	Variable†	Therapy started during PCI and continued for 6–24 hours thereafter
Nitrates: Short acting		
Sublingual nitroglycerin (tablet or spray)	0.3–0.6 mg every 4–5 minutes up to 3 doses	All patients for immediate relief of chest pain; used as needed
Nitroglycerin as continuous IV drip	Started at 5 mcg/minute and increased 2.5–5.0 mcg	Selected patients with ACS:

Medication	Dosage	Use
	every few minutes until required response occurs	During the first 24 to 48 hours, those with heart failure (unless hypotension is present), large anterior myocardial infarction, persistent angina, or hypertension (BP is reduced by 10–20 mm Hg but not to < 80–90 mm Hg systolic) For longer use, patients with recurrent angina or persistent pulmonary congestion
Nitrates: Long acting		
Isosorbide dinitrate	10–20 mg orally 3 times a day; can be increased to 40 mg 3 times a day	Patients who have unstable angina or persistent severe angina and continue to have anginal symptoms after the beta-blocker dose is maximized A nitrate-free period of about 8–10 hours (typically at night) recommended to avoid tolerance (specific drugs recommend different durations of nitrate-free period)
Isosorbide dinitrate (sustained-release)	40–80 mg orally twice a day (typically given at 8 AM and 2 PM)	
Isosorbide mononitrate	20 mg orally twice a day, with 7 hours between 1st and 2nd doses	
Isosorbide mononitrate (sustained-release)	30 or 60 mg orally once a day, increased to 120 mg or, rarely, 240 mg	
Nitroglycerin patches	0.2–0.8 mg/hour applied between 6:00 and 9:00 AM and removed 12–14 hours later to avoid tolerance	
Nitroglycerin ointment 2% preparation (15 mg/2.5 cm)	1.25 cm spread evenly over upper torso or arms every 6 to 8 hours and covered with plastic, increased to 7.5 cm as tolerated, and removed	

Medication	Dosage	Use
	for 8–12 hours each day to avoid tolerance	
Opioids		
Morphine	2–4 mg IV, repeated as needed	Morphine should be used judiciously (eg, if nitroglycerin is contraindicated or if patient has symptoms despite maximal doses of nitroglycerin) given a possible increase in mortality as well as attenuation of P2Y12 receptor inhibitor activity
PCSK-9 inhibitors		
Alirocumab	Initial dose: 75 mg subcutaneously, once every 2 weeks or 300 mg subcutaneously once every 4 weeks	For patients not at target LDL-C levels, used alone or in combination with other lipid-lowering therapies (eg, statins, ezetimibe) for the treatment of adults with primary hyperlipidemia (including familial hypercholesterolemia)
Evolocumab	Initial dose for primary hyperlipidemia: 140 mg subcutaneously every 2 weeks or 420 mg subcutaneously once monthly	For patients not at target LDL-C levels, used alone or in combination with other lipid-lowering therapies (eg, statins, ezetimibe) for the treatment of adults with primary hyperlipidemia (including familial hypercholesterolemia)
Statins (HMG-CoA reductase inhibitors)		
Atorvastatin Fluvastatin	Variable† (see Dyslipidemia)	Patients with CAD should be given maximally

Medication	Dosage	Use
Lovastatin Pravastatin Rosuvastatin Simvastatin		tolerated statin dose
Other medications		
Ivabradine	5 mg orally twice a day, increased to 7.5 mg orally twice a day if needed	Inhibits sinus node For symptomatic treatment of chronic stable angina pectoris in patients with normal sinus rhythm who cannot take beta-blockers In combination with beta-blockers in patients inadequately controlled by beta-blocker alone and whose heart rate > 60 beats/minute
Ranolazine	500 mg orally twice a day, increased to 1000 mg orally twice a day as needed	Patients in whom anginal symptoms continue despite treatment with other antianginal drugs

* Clinicians may use different combinations of medications depending on the type of coronary artery disease that is present.

† Refer to the manufacturer's prescribing information or a drug information database for specific dosing information.

‡ Lower doses of Direct-acting oral anticoagulants are sometimes used, based on patient-specific risk factors.

§ Higher doses of aspirin do not provide greater protection and increase risk of adverse effects.

¶ Of low molecular weight heparins (LMWHs), enoxaparin is preferred.

ACS = acute coronary syndromes; BP = blood pressure; CABG = coronary artery bypass grafting; CAD = coronary artery disease; HMG = hydroxymethylglutaryl; INR = international normalized ratio; LDL-C = low-density lipoprotein cholesterol; LV = left ventricular; MI = myocardial infarction; NSTEMI = non–ST-segment elevation MI; PCI = percutaneous intervention; STEMI = ST-segment elevation MI.

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Antiplatelet Agents

Aspirin, clopidogrel, prasugrel, ticagrelor, and glycoprotein (GP) IIb/IIIa inhibitors are examples of antiplatelet agents. In general, the goal with dosage and duration of antiplatelet agent use is to balance the decreased risk of coronary thrombosis with the increased risk of bleeding.

All patients are given aspirin 160 to 325 mg (not enteric-coated), if not contraindicated (eg, life-threatening active bleeding). Chewing the first dose before swallowing quickens absorption. Aspirin reduces short- and long-term mortality risk (1).

If aspirin cannot be taken initially, clopidogrel 75 mg orally once a day or ticlopidine 250 mg orally twice a day may be used. Clopidogrel has largely replaced ticlopidine for routine use because of the risk of neutropenia with ticlopidine.

Patients not undergoing revascularization

Patients with acute coronary syndrome (ACS) in whom intervention is not possible or recommended are treated with dual antiplatelet therapy with aspirin and a P2Y12 inhibitor (ticagrelor, prasugrel, or clopidogrel) for at least 12 months before transitioning to aspirin monotherapy. A proton pump inhibitor should be given with dual antiplatelet therapy.

Patients undergoing revascularization

In patients undergoing PCI, a loading dose of clopidogrel (300 to 600 mg orally once), prasugrel (60 mg orally once), or ticagrelor (180 mg orally once) improves outcomes (1). For urgent PCI, prasugrel and ticagrelor are more rapid in onset and may be preferred. IV cangrelor may also be used during PCI to reduce ischemic events in patients who are P2Y12-inhibitor naive.

Some clinicians give a GP IIb/IIIa inhibitor during (or sometimes before) PCI for patients with high-risk lesions (high thrombus burden, inadequate perfusion despite relief of obstruction ["no reflow"]) (2). The GP IIb/IIIa inhibitor is continued for 6 to 24 hours, and angiography is performed before the infusion period is over. GP IIb/IIIa inhibitors are not recommended for patients receiving fibrinolytics. Abciximab, tirofiban, and eptifibatide appear to have equivalent efficacy, and the choice of agent should depend on other factors (eg, cost, availability, familiarity).

After PCI, patients may receive dual antiplatelet therapy as for patients who are not undergoing revascularization. However, if bleeding risk is a concern, patients can be transitioned to monotherapy with aspirin or a P2Y12 inhibitor after 1 month, or de-escalated from ticagrelor or prasugrel to clopidogrel, which has a lower bleeding risk, as part of ongoing dual antiplatelet therapy. Thereafter,the use of P2Y12 inhibitors as an option for long term antiplatelet monotherapy after the discontinuation of dual antiplatelet therapy represents a change from the historical practice of using long-term low-dose aspirin.

Patients requiring long-term anticoagulation (eg, patients with atrial fibrillation) who are on "triple" therapy (dual antiplatelet plus an anticoagulant) after PCI can be transitioned to clopidogrel antiplatelet monotherapy along with their oral anticoagulant after 1 to 4 weeks.

Antiplatelet agent references

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Anticoagulants

Either a low molecular weight heparin (LMWH), unfractionated heparin, or bivalirudin is given routinely to patients with acute coronary syndrome unless contraindicated (eg, by active bleeding or planned use of streptokinase or anistreplase) (1).

Choice of agent depends on whether revascularization is anticipated, as well as other factors such as patient characteristics, cost, and local practice. The LMWHs have better bioavailability, are given by simple weight-based dose without monitoring activated partial thromboplastin time (aPTT) and dose titration, and have lower risk of heparin-induced thrombocytopenia; they are recommended when an early invasive strategy is not planned. Bivalirudin is recommended for patients with a known or suspected history of heparin-induced thrombocytopenia who are undergoing PCI. Anticoagulants are continued for:

- Duration of PCI in patients undergoing this procedure
- Duration of hospital stay (in patients on LMWH) or 48 hours (in patients on unfractionated heparin) in all other cases

Patients at high risk of systemic emboli (eg, atrial fibrillation with CHA2DS2VASc score ≥ 2) also require long-term therapy with an oral anticoagulant (eg, warfarin, dabigatran, apixaban, rivaroxaban). Conversion to oral anticoagulants should begin 48 hours after symptom resolution or PCI.

Unfractionated heparin

Unfractionated heparin is a recommended therapy for use in patients with non-ST-segment elevation acute coronary syndrome (NSTE-ACS), and in patients with any ACS undergoing PCI (1).

Unfractionated heparin is more complicated to use because it requires frequent (every 6 hours) dosing adjustments to achieve an aPTT 1.5 to 2 times the control value. In patients undergoing angiography, further dosing adjustment is made to achieve an activated clotting time (ACT) of 200 to 250 seconds if the patient is treated with a GP IIb/IIIa inhibitor and 250 to 300 seconds if a GP IIb/IIIa inhibitor is not being given. However, if bleeding develops after catheterization, the effects of unfractionated heparin are shorter and can be reversed (by promptly stopping the heparin infusion and giving protamine sulfate).

Low molecular weight heparin

Enoxaparin is a recommended alternative to unfractionated heparin therapy in patients with NSTE-ACS in whom an early invasive strategy is not planned, and IV enoxaparin may be considered as an alternative therapy to unfractionated heparin or bivalirudin in patients with ACS undergoing PCI (1). Enoxaparin is also the recommended therapy for patients undergoing fibrinolysis.

The LMWHs have better bioavailability, are given by simple weight-based dosing without monitoring aPTT and dose titration, and have lower risk of heparin-induced thrombocytopenia. They also may produce an incremental benefit in outcomes relative to unfractionated heparin in patients with ACS. Of the LMWHs, enoxaparin is the recommended agent (1). However, enoxaparin dosing is adjusted in patients > 75 years old with STEMI due to a higher bleeding risk (2), and its effects are not completely reversible with protamine.

Heparin alternatives

The direct thrombin inhibitor **bivalirudin** has a lower incidence of serious bleeding and improved outcomes compared to heparin (1). Bivalirudin is an acceptable alternative to unfractionated heparin for patients undergoing PCI who are at high risk of bleeding.

The factor Xa inhibitor **fondaparinux** is a recommended alternative therapy to unfractionated heparin in patients with NSTE-ACS in whom an early invasive strategy is not planned and is a recommended alternative therapy to enoxaparin in patients undergoing fibrinolytic therapy. Fondaparinux is not recommended in patients undergoing PCI; it has been shown to cause a higher rate of complications without benefit over unfractionated heparin in patients with STEMI undergoing PCI (3).

Oral anticoagulants

For patients who require anticoagulation for another reason (eg, atrial fibrillation), direct-acting oral anticoagulants (apixaban, dabigatran, edoxaban, rivaroxaban) are preferred long term over warfarin, unless there is a contraindication to them (1). For most patients requiring anticoagulation, triple therapy with oral anticoagulation, a P2Y12 inhibitor, and aspirin is continued up to 1 week to 1 month after intervention, and then patients are continued on oral anticoagulation and a P2Y12 inhibitor for 6 months to 1 year. Calcium channel blockers and nitrates may also be given to reduce risk of coronary spasm.

Anticoagulants references

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Beta-Blockers

Beta-blockers are recommended unless contraindicated (eg, by bradycardia, heart block, hypotension, or asthma), especially for high-risk patients. Beta-blockers reduce heart rate, arterial pressure, and contractility, thereby reducing cardiac workload and oxygen demand. Infarct size largely determines cardiac performance after recovery. Oral beta-blockers given within the first few hours improve prognosis by reducing infarct size, recurrence rate, incidence of ventricular fibrillation, and mortality risk (1), but they have limited benefit and are not routinely used in patients with preserved left ventricular ejection fraction (2, 3, 4).

Heart rate and BP must be carefully monitored during treatment with beta-blockers. Dosage is reduced if bradycardia or hypotension develops. Excessive adverse effects may be reversed by infusion of the beta-adrenergic agonist isoproterenol 1 to 5 mcg/minute.

Beta-blocker references

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Nitrates

A short-acting nitrate, nitroglycerin, is used to reduce pain and cardiac workload in selected patients. Nitroglycerin dilates veins, arteries, and arterioles, reducing left ventricular preload and afterload. As a result, myocardial oxygen demand is reduced, lessening ischemia.

Chest pain can be treated with sublingual or intravenous nitroglycerin, or sometimes morphine or fentanyl. Nitroglycerin is preferable to opioids (1), which should be used judiciously (eg, if a patient has a contraindication to nitroglycerin or is in pain despite maximal nitroglycerin therapy). Nitroglycerin is initially given sublingually, followed by continuous IV drip if needed for pain, hypertension, or pulmonary edema. Nitroglycerin in either form should not be used if the systolic blood pressure is below 90 mm Hg or if the decrease in systolic blood pressure is > 30 mm Hg from baseline.

Nitroglycerin has been used historically with the intent of reducing infarct size and improve outcomes, but large trials found no evidence of benefit and this practice is no longer recommended (2, 3).

Nitrates references

1. Rao SV, O'Donoghue ML, Ruel M, et al. 2025 ACC/AHA/ACEP/NAEMSP/SCAI Guideline for the Management of Patients With Acute Coronary Syndromes: A Report of the American College of Cardiology/American Heart Association Joint Committee on Clinical Practice Guidelines. *Circulation.* 2025;151(13):e771-e862. doi:10.1161/CIR.0000000000001309
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Fibrinolytics

Tenecteplase (TNK), alteplase (rTPA), reteplase (rPA), and streptokinase, all given IV, are plasminogen activators. They convert single-chain plasminogen to double-chain plasminogen, which has fibrinolytic activity. They have different characteristics and dosing regimens (see table IV Fibrinolytic Therapies) and are appropriate only for selected patients with STEMI.

Tenecteplase and reteplase are recommended most often because of their simplicity of administration; tenecteplase is given as a single bolus over 5 seconds and reteplase as a double bolus 30 minutes apart; alteplase is given as an infusion over 90 minutes (1). Tenecteplase, reteplase, and alteplase are preferred over other fibrinolytics. Reteplase appears to have higher recanalization rates than alteplase with no difference in bleeding (2).; Tenecteplase has been shown to be noninferior, and possibly superior when combined with ticagrelor to alteplase in terms of recanalization (3, 4).

Streptokinase (not available in the United States) may induce allergic reactions, especially if it has been used previously, and must be given by infusion over 30 to 60 minutes; however, it has a low incidence of intracerebral hemorrhage and is relatively inexpensive. Anistreplase, related to streptokinase, is similarly allergenic and slightly more expensive but can be given as a single bolus. Recanalization rate is lower than that with other plasminogen activators. Because of the possibility of allergic reactions, patients who previously received streptokinase or anistreplase are not given that medication again.

TABLE

IV Fibrinolytic Therapies

Medication	Circulating half-life (minutes)	Concurrent heparin	Allergic reactions
Alteplase	6	Yes	Rare
Reteplase	13–16	Yes	Rare
Streptokinase (not available in the	20	No	Yes

Medication	Circulating half-life (minutes)	Concurrent heparin	Allergic reactions
United States)			
Tenecteplase	Initial half-life of 20–24 minutes; terminal phase half-life of 90–130 minutes	Yes	Rare

CLINICAL CALCULATORS

Intracranial Bleeding Risk from Thrombolytic Therapy of MI

Contraindications to fibrinolytic therapy

There are many absolute and relative contraindications to fibrinolytic therapy. In general, the presence of active bleeding or a condition where bleeding would be life-threatening is an absolute contraindication (see Table Contraindications to Fibrinolytic Therapy for STEMI).

Fibrinolytics references

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Other Medications for Post-acute Treatment of Acute Coronary Syndrome

Angiotensin-converting enzyme (ACE) inhibitors and **angiotensin II receptor blockers** (ARBs) reduce mortality risk in patients with myocardial infarction, especially in those with anterior infarction, left ventricular ejection fraction ≤ 40%, hypertension, or diabetes (1). The greatest benefit occurs in the patients at highest risk early during convalescence. ACE inhibitors and ARBs are given > 24 hours after thrombolysis stabilization and, because of continued beneficial effect, may be prescribed long-term. They provide long-term cardioprotection by improving endothelial function (2), particularly in patients with heart failure, hypertension, diabetes, or chronic kidney disease (3). Contraindications include hypotension, kidney failure, bilateral renal artery stenosis, and known allergy.

Mineralocorticoid receptor antagonists (spironolactone or eplerenone) are indicated to reduce morbidity and mortality in patients with left ventricular dysfunction and either heart failure symptoms or diabetes following ACS (1).

In patients with heart failure, data also support the use of **sodium–glucose co-transporter 2 (SGLT2) inhibitors** after MI, regardless of diabetes status, to reduce the risk of worsening heart failure, cardiovascular mortality, or both (1, 3).

Statins (HMG-CoA reductase inhibitors) have long been used for prevention of coronary artery disease and acute coronary syndromes, but there is evidence that they also have short-term benefits, such as stabilizing plaque, reversing endothelial dysfunction, decreasing thrombogenicity, and reducing inflammation. Thus, all patients without contraindications (eg, statin-induced myopathy, liver dysfunction) to therapy should receive a statin at the maximally tolerated dose as early as possible following ACS regardless of their serum lipid levels (1, 4).

PCSK-9 inhibitors (evolocumab, alirocumab), the **cholesterol absorption inhibitor** ezetimibe, and bempedoic acid are used for patients not at target low-density lipoprotein cholesterol levels on maximal statin therapy alone or for those who cannot tolerate a statin (1, 5).

In patients who have had an MI, icosapent ethyl is used in patients with hypertriglyceridemia to reduce risk of subsequent ischemic events and cardiovascular death (6).

Long-term **colchicine** appears to improve outcomes in patients with chronic coronary disease, after MI, and possibly after ACS in general. (1, 7).

Other medications references

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Drug Information for the Topic

Complications of Acute Coronary Syndromes

*Full Review: Jul 2026 By **Ranya N. Sweis**, MD, MS, Northwestern University Feinberg School of Medicine | **Arif Jivan**, MD, PhD, Northwestern University Feinberg School of Medicine | Peer reviewed by **Jonathan G. Howlett**, MD, Cumming School of Medicine, University of Calgary*
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Numerous complications can occur as a result of an acute coronary syndrome and increase morbidity and mortality. Complications can be roughly categorized as:

- Electrical dysfunction (ventricular and atrial arrhythmias, sudden cardiac death, and sinus node and conduction disturbances)
- Mechanical dysfunction (hypotension or shock, heart failure, myocardial rupture or aneurysm, papillary muscle dysfunction)
- Thrombotic complications (recurrent coronary ischemia, mural thrombosis)
- Inflammatory complications (pericarditis, post-myocardial infarction syndrome)

Complications are not uncommon after acute myocardial infarction (MI). Females have higher rates of mechanical complications than males, whereas rates are similar for ventricular arrhythmias (1).

General reference

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Electrical Dysfunction

Electrical dysfunction has been reported to occur in up to 90% of patients after myocardial infarction (MI) (see also Arrhythmias and Conduction Disorders) (1). Ventricular tachycardia or ventricular fibrillation occurs within 24 hours after percutaneous coronary intervention (PCI) for ST-segment elevation myocardial infarction (STEMI) in approximately 9% of patients (2). Electrical disturbances associated with increased mortality include second-degree or third-degree atrioventricular (AV) block, ventricular tachycardia, ventricular fibrillation, and atrial fibrillation (3, 4). Asystole is uncommon, except as a terminal manifestation of progressive left ventricular failure and shock. Patients with disturbances of cardiac rhythm are evaluated for hypoxia and electrolyte abnormalities, which can be causative or contributory.

Ventricular Arrhythmias

Ventricular arrhythmias are common and may result from hypoxia, electrolyte imbalance (hypokalemia, possibly hypomagnesemia), or sympathetic overactivity in ischemic cells adjacent to infarcted tissue (which is not electrically active). Treatable causes of ventricular arrhythmias are sought and corrected.

Serum potassium should be kept above 4.0 mEq/L (4.0 mmol/L). IV potassium chloride is recommended; usually 10 mEq/hour (10 mmol/hour) can be infused, but for severe hypokalemia (potassium level < 2.5 mEq/L [2.5 mmol/L]), 20 to 40 mEq/hour (20 to 40 mmol/hour) can be infused through a central venous line.

During the first 40 days after MI, beta-blockers and sometimes other antiarrhythmic therapy are used to prevent and treat ventricular arrhythmias (5). Early use of oral beta-blockers reduces the incidence of ventricular arrhythmias (including ventricular fibrillation) and mortality in patients who do not have heart failure or hypotension (6), although IV beta-blockade has been associated with an increased risk of cardiogenic shock (7). Prophylaxis with other medications (eg, lidocaine) may increase mortality risk and is not recommended (8). The role of implantable cardioverter-defibrillators (ICD) placed between 6 and 40 days after MI in patients with significant ventricular arrhythmias is unclear.

After the acute phase (≥ 40 days after MI and 90 days after revascularization), the presence of complex ventricular arrhythmias or nonsustained ventricular tachycardia, especially with significant left ventricular systolic dysfunction, increases mortality risk. Programmed endocardial stimulation can help select the most effective antiarrhythmics or determine the need for an implantable cardioverter-defibrillator (ICD). An ICD should be considered and is indicated for the following groups of patients when life expectancy is at least 1 year:

- All patients with left ventricular ejection fraction (LVEF) ≤ 40% with inducible ventricular tachycardia
- Patents with LVEF 31 to 35% and New York Heart Association class II or class III heart failure
- Patients with LVEF ≤ 30% and New York Heart Association class I, II, or III heart failure

Before treatment with an antiarrhythmic or ICD, coronary angiography and other tests are performed to look for recurrent myocardial ischemia, which may require PCI or coronary artery bypass grafting (CABG).

Ventricular ectopic beats, which are common after myocardial infarction, do not warrant specific treatment.

Ventricular tachycardia

Nonsustained ventricular tachycardia (ie, < 30 seconds) and even sustained slow ventricular tachycardia (accelerated idioventricular rhythm) without hemodynamic instability do not usually require treatment in the first 24 to 48 hours (see figure Broad QRS Ventricular Tachycardia).

Synchronized cardioversion is performed for:

- Polymorphic ventricular tachycardia
- Sustained (≥ 30 seconds) monomorphic ventricular tachycardia

- Any ventricular tachycardia with symptoms of instability (eg, heart failure, hypotension, chest pain)

Ventricular tachycardia without hemodynamic instability may be treated with IV lidocaine, procainamide, or amiodarone. Some clinicians also treat complex ventricular arrhythmias with magnesium sulfate 2 g IV over 5 minutes whether or not serum magnesium level is low.

Ventricular tachycardia may occur months after myocardial infarction. Late ventricular tachycardia is more likely to occur in patients with transmural infarction and to be sustained.

Broad QRS Ventricular Tachycardia

The QRS duration is 160 millisecond. An independent P wave can be seen in II (*arrows*). There is a leftward mean frontal axis shift.

Ventricular fibrillation

Ventricular Fibrillation

IMAGE

These rhythm strips show the ultrarapid baseline undulations, irregular in timing and morphology, that characterize ventricular fibrillation. The asterisks indicate close-coupled premature ventricular beats.

Ventricular fibrillation occurs in some patients during the first 24 hours after myocardial infarction, usually within 6 hours (2). Late ventricular fibrillation usually indicates continued or recurrent myocardial ischemia and, when accompanied by hemodynamic deterioration, is a poor prognostic sign. Ventricular fibrillation is treated with immediate unsynchronized cardioversion (defibrillation).

Sinus Node Disturbances

If the artery supplying the sinus node is affected by an acute coronary syndrome, sinus node disturbances can occur; they are more likely if there is a preexisting sinus node disorder (common among older patients).

Sinus bradycardia

Sinus bradycardia, the most common sinus node disturbance, is usually not treated unless there is hypotension or the heart rate is < 50 beats/minute. A lower heart rate, if not extreme, means reduced cardiac workload and possibly reduced infarct size.

For bradycardia with hypotension (which may reduce myocardial perfusion), atropine 0.5 to 1 mg IV is used; it can be repeated after several minutes if response is inadequate. Several small doses are best because high doses may induce tachycardia. Occasionally, a temporary or even permanent transvenous pacemaker must be inserted (9).

Sinus tachycardia

Persistent sinus tachycardia is usually ominous, often reflecting left ventricular failure and low cardiac output. Without heart failure or another evident cause, this arrhythmia may respond to a beta-blocker, given orally or intravenously depending on degree of urgency.

Atrial Arrhythmias

Atrial arrhythmias (atrial ectopic beats, atrial fibrillation, and, less commonly, atrial flutter) occur in approximately 10 to 20% of patients who have had a myocardial infarction and may reflect left ventricular failure or right atrial infarction (3, 10). Atrial fibrillation is more common after non–ST-segment elevation myocardial infarction (NSTEMI) than STEMI.

Paroxysmal atrial tachycardia usually occurs in patients who have had previous episodes of it.

Atrial ectopy is usually benign, but if frequency increases, causes, particularly heart failure, are sought. Frequent atrial ectopic beats may respond to a beta-blocker.

Atrial fibrillation

Atrial fibrillation is usually transient if it occurs within the first 24 hours (see figure Atrial Fibrillation). Risk factors for developing atrial fibrillation after MI include female gender, older age, increased heart rate on admission, and decreased LVEF (11, 12, 13).

Atrial Fibrillation

Recurrent paroxysmal atrial fibrillation is a poor prognostic sign and increases risk of systemic emboli (4, 14).

For atrial fibrillation, an anticoagulant is usually given because of the risk of systemic emboli. Patients are given an oral anticoagulant, along with aspirin and a P2Y12 inhibitor ("triple therapy"). These medications are continued for 1 to 4 weeks before the patient is transitioned to the oral anticoagulant and P2Y12 inhibitor for 12 months (5, 15).

Intravenous beta-blockers (eg, atenolol 2.5 to 5.0 mg over 2 minutes to total dose of 10 mg in 10 to 15 minutes, metoprolol 2 to 5 mg every 2 to 5 minutes to a total dose of 15 mg in 10 to 15 minutes) rapidly slow the ventricular rate and are typically given when heart rate is > 100 beats per minute. Heart rate and blood pressure are closely monitored. Treatment is withheld when ventricular rate decreases satisfactorily, systolic blood pressure is < 100 mm Hg, or significant bronchospasm occurs (10).

Other options for rate control include digoxin, non-dihydropyridine calcium channel blockers, and amiodarone (10).

Intravenous digoxin, which is not as effective as beta-blockers, is used cautiously and only in patients with atrial fibrillation and left ventricular systolic dysfunction. Usually, digoxin takes at least 2 hours to effectively slow heart rate and may rarely aggravate ischemia in patients with recent acute coronary syndrome.

For patients without evident left ventricular systolic dysfunction or conduction delay manifested by a wide QRS complex, the non-dihydropyridine IV calcium channel blockers verapamil or diltiazem may be used for rate control when beta-blockers are contraindicated or if adequate ventricular rate control is not achieved with other agents. Diltiazem may be given as a continuous IV infusion to control heart rate for long periods.

Intravenous amiodarone may also be used for treatment of acute atrial fibrillation, especially when IV beta-blockade or calcium channel blockade is not appropriate or contraindicated (such as in patients with low blood pressure or active asthma).

Due to a high risk of recurrent atrial fibrillation in patients with acute MI, an initial cardioversion strategy may not be preferred. However, if atrial fibrillation compromises circulatory status (eg, causing left ventricular failure, hypotension, or chest pain), urgent electrical synchronized cardioversion is performed (10). If atrial fibrillation returns after cardioversion, IV amiodarone should be considered for patients who continue to experience symptoms (eg, chest pain) or remain hemodynamically compromised.

Atrial flutter

For atrial flutter (see also figure Atrial Flutter), rate is controlled as for atrial fibrillation; anticoagulation along with antiplatelet therapy is required because the risk of thromboembolism is similar to that with atrial fibrillation. Rate control for atrial flutter in patients with acute MI is usually unsatisfactory. Low-energy direct current (DC) synchronized cardioversion will usually terminate atrial flutter.

Atrial Flutter

(Note: Conducted with right bundle branch block.)

Conduction Defects

Mobitz type I block (Wenckebach block, progressive prolongation of the PR interval with eventual dropped beats) is relatively common with an inferior-diaphragmatic infarction (see figure Mobitz Type I Second-Degree Atrioventricular Block); it is usually self-limited and rarely progresses to higher grade block.

Classic Mobitz Type I Second-Degree Atrioventricular Block

The PR interval progressively lengthens with each beat until the atrial impulse is not conducted and the QRS complex is dropped (Wenckebach phenomenon); atrioventricular nodal conduction resumes with the next beat, which has the shortest PR interval, and the sequence is repeated.

Mobitz type II block (dropped beats without progressive lengthening of the PR interval) usually indicates massive anterior myocardial infarction and is associated with an increased mortality risk, as does complete heart block with wide QRS complexes (atrial impulses do not reach the ventricle); both are uncommon (5, 16, 17). Complete AV block (see figure Third-Degree Atrioventricular Block) occurs in 2 to 10% of patients with after myocardial infarction (higher in non-anterior locations).

Frequency of third-degree atrioventricular block (complete block) depends on site of infarction, and its presence confers an increased risk of mortality (16, 17).

Third-Degree Atrioventricular Block

Treatment of heart block following MI

Mobitz type I block usually does not warrant treatment.

For true Mobitz type II block with dropped beats or for AV block with slow, wide QRS complexes, temporary transvenous pacing is the treatment of choice. External pacing can be used until a temporary transvenous pacemaker can be placed. A permanent pacemaker is required for patients with third-degree block and those with persistent second-degree AV block, especially if symptomatic.

Although isoproterenol infusion may restore rhythm and rate temporarily, it is not used because it increases oxygen demand and the risk of rhythm abnormalities. IV atropine (eg, 0.5 mg IV every 3 to 5 minutes to a total dose of 3 mg) may be useful for narrow-complex atrioventricular block with a slow ventricular rate but is not recommended for new wide-complex atrioventricular block (indicating conducting system involvement disease below the level of the AV node) (12).

High-grade AV block refractory to appropriate medical therapy should generally be treated with temporary pacing, with permanent pacing considered in most cases if the conduction system has not recovered after 72 hours (5, 9).

Electrical dysfunction references

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Hypotension, Cardiogenic Shock, and Heart Failure

Hypotension

Hypotension may be due to:

- Decreased ventricular filling
- Loss of contractile force secondary to massive myocardial infarction

Decreased left ventricular filling is most often caused by reduced venous return secondary to hypovolemia, especially in patients receiving intensive loop diuretic therapy, but it may reflect right ventricular infarction. Marked pulmonary congestion suggests loss of left ventricular contractile force (left ventricular failure) as the cause.

Treatment depends on the cause. In some patients, determining the cause requires use of a pulmonary artery catheter to measure intracardiac pressures.

For hypotension due to hypovolemia, cautious fluid replacement with 0.9% saline is usually possible without left heart overload (excessive rise in left atrial pressure). However, sometimes left ventricular function is so compromised that adequate fluid replacement sharply increases pulmonary artery occlusion pressure to levels associated with pulmonary edema (> 25 mm Hg). If left atrial pressure is high, hypotension is probably due to left ventricular failure, and if diuretics are ineffective, inotropic therapy or mechanical circulatory support may be required.

Right Ventricular Ischemia or Infarction

Right ventricular infarction rarely occurs in isolation; it usually accompanies inferior left ventricular infarction. The first sign may be hypotension developing in a patient whose condition had been previously stable.

Right-sided ECG leads may show ST-segment changes. Volume loading with 1 to 2 L of 0.9% saline is often effective. Dobutamine or milrinone (which has better dilating effects on the pulmonary circulation) may help. Nitrates and diuretics are not used; they reduce preload (and hence cardiac output), causing severe hypotension. Increased right-sided filling pressure should be maintained by IV fluid infusion, but excessive volume overload may compromise left ventricular filling and cardiac output.

Pearls & Pitfalls

- With hypotension after myocardial infarction or inferior STEMI, do ECG with right-sided leads to diagnose right ventricular infarction, and, if confirmed, avoid nitrates and diuretics.

Cardiogenic Shock

Approximately 5 to 10% of patients with acute myocardial infarction have cardiogenic shock (1).

Marked hypotension (eg, systolic blood pressure < 90 mm Hg) with tachycardia and symptoms of end-organ hypoperfusion (reduced urine output, mental confusion, diaphoresis, cold extremities) is termed cardiogenic shock. Pulmonary congestion develops rapidly in cardiogenic shock.

Definitive treatment for cardiogenic shock caused by acute MI is revascularization of the specific vessel involved by percutaneous coronary intervention (PCI) or coronary artery bypass grafting (CABG) if PCI is not feasible (2). Revascularization usually greatly improves ventricular function. Treatment with PCI of stenotic arteries not related to the infarct causing cardiogenic shock is deferred.

Supportive care often involves inotropic support (eg, dobutamine, milrinone, epinephrine, norepinephrine, levosimendan), with the inotropic effects of any agent balanced carefully against chronotropic effects, which can affect myocardial oxygen demand, and vasoconstrictor effects, which can affect systemic afterload.

Temporary mechanical circulatory support can be provided with a microvascular intra-axial flow pump, which unloads the left ventricle by pumping blood directly into the aorta and can be placed percutaneously or surgically (2). Other technologies, such as an intraortic balloon pump or veno-arterial extracorporeal membrane oxygenation, are not recommended. Temporary mechanical circulatory support can allow the myocardium to recover or serve as a bridge toward long-term mechanical support or heart transplantation.

Heart Failure

Heart failure occurs in 14 to 36% of patients hospitalized with MI (3). Heart failure following MI is more likely in patients with (4):

- Chronic kidney disease
- Recurrent MI
- African American ancestry
- Diabetes

The extent of myocardial dysfunction, overall hemodynamic status, and clinical findings depend on infarct size, elevation of left ventricular filling pressure, and degree of reduction in cardiac output. Dyspnea, inspiratory crackles at the lung bases, and hypoxemia are common.

Initial treatment depends on severity.

For mild cases, a loop diuretic (eg, furosemide 20 to 40 mg IV once or twice a day) to reduce ventricular filling pressure is often sufficient. For severe cases, vasodilators (eg, IV nitroglycerin, nitroprusside) are often used to reduce preload and afterload; these agents are effective acutely (eg, in acute pulmonary edema) and may be continued over 24 to 72 hours as necessary. During treatment, pulmonary artery occlusion pressure may be measured via right heart (pulmonary artery) catheterization, especially if the response to therapy is not as desired.

Longer term heart failure therapy is added as indicated, which can include renin-angiotensin system inhibitors (or angiotensin receptor/neprilysin inhibitor), beta-blockers, mineralocorticoid receptor antagonists, and sodium-glucose cotransporter-2 inhibitors. (See Treatment of Chronic Heart Failure.) Note that in the post-MI setting, the benefit of universal beta-blockade in patients with LVEF ≥ 40% is equivocal (5, 6).

For severe heart failure, a microvascular intra-axial flow pump may provide temporary hemodynamic support until the patient's condition stabilizes or the decision is made to provide more advanced support (2). When revascularization or surgical repair is not feasible, heart transplantation is considered. Long-term left ventricular or biventricular implantable assist devices may be used as a bridge to transplantation. If transplantation is not possible, the left ventricular assist device may be used as permanent treatment (destination therapy). Occasionally, use of such a device results in recovery, and the device can be removed in 3 to 6 months.

Hypotension, cardiogenic shock, and heart failure references

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Mechanical Dysfunction

Generally, mechanical complications should be managed in a cardiac surgical facility (1). Temporary mechanical circulatory support may be necessary until the complication can be addressed surgically.

Papillary Muscle Disorders

Functional **papillary muscle insufficiency,** as detected by late gadolinium enhancement on cardiac MRI, occurs in up to 40% of patients during the first few hours of infarction (2). Papillary muscle ischemic dysfunction causes incomplete coaptation of the mitral valve leaflets, which is transient in most patients. But in some patients, papillary muscle or free wall scarring causes permanent mitral regurgitation. Functional papillary muscle insufficiency is characterized by an apical late systolic murmur and typically resolves without treatment.

Papillary muscle rupture occurs most often after an inferoposterior infarct due to right or circumflex coronary artery occlusion that affects the posteromedial papillary muscle, which has a single blood supply (unlike the anterolateral papillary muscle, which has a dual blood supply) (3). It causes acute, severe mitral regurgitation. Papillary muscle rupture is characterized by the sudden appearance of a loud apical holosystolic murmur and thrill, usually with pulmonary edema. Occasionally, severe regurgitation is silent. An abrupt hemodynamic deterioration raises clinical suspicion of papillary muscle rupture; echocardiography should be performed to make the diagnosis. Urgent mitral valve repair or replacement is necessary and effective.

Myocardial Rupture

Interventricular septum or free wall rupture occurs in 1% of patients with acute myocardial infarction. It causes 15% of in-hospital mortality.

Interventricular septum rupture occurs in approximately 0.3% of patients; is more common in females, older patients, and those who have had delayed reperfusion; and confers an in-hospital mortality of approximately 40% even with surgery (3). Interventricular septum rupture is characterized by the sudden appearance of a loud systolic murmur and thrill medial to the apex along the left sternal border in the third or fourth intercostal space, which is accompanied by hypotension with or without signs of left ventricular failure. Diagnosis may be confirmed using a balloon-tipped catheter and comparing blood oxygen saturation or partial pressure of oxygen (PO_2) of right atrial, right ventricular, and pulmonary artery samples. A significant increase in right ventricular PO_2 is diagnostic, as is Doppler echocardiography, which may demonstrate the actual shunt of blood across the ventricular septum.

Treatment is surgery, which should be delayed if possible for up to 6 weeks after MI so that infarcted myocardium can heal maximally; if hemodynamic instability persists, earlier surgery is indicated despite a high mortality risk. Patients with high surgical risk may undergo transcatheter closure.

Free wall rupture is thought to be an important cause of out-of-hospital sudden cardiac death; its incidence is not known (3). It is characterized by sudden loss of arterial pressure with momentary persistence of sinus rhythm and often by signs of cardiac tamponade. Rupture of a free wall is almost always fatal. When discovered in time, surgery can be successful, but mortality after surgery is approximately 35%.

Ventricular Aneurysm and Pseudoaneurysm

A localized bulge in the ventricular wall, usually the left ventricular anterior or apical wall, can occur at the site of a large infarction (3). Ventricular aneurysms are common, especially with a large transmural infarct (usually anterior). Aneurysms may develop in a few days, weeks, or months. They are unlikely to rupture but may lead to recurrent ventricular arrhythmias, low cardiac output, and mural thrombosis with systemic embolism.

A ventricular aneurysm may be suspected when paradoxical precordial movements are seen or felt, ECG shows persistent ST-segment elevation, and chest radiograph shows a characteristic bulge of the cardiac shadow. Because these findings are not diagnostic of an aneurysm, echocardiography is performed to confirm the diagnosis and determine whether a thrombus is present.

Surgical excision may be indicated when left ventricular failure or arrhythmia persists. Early revascularization and probably the use of angiotensin-converting enzyme (ACE) inhibitors during acute myocardial infarction modify left ventricular remodeling and have reduced the incidence of aneurysm.

Pseudoaneurysm is incomplete rupture of the free left ventricular wall (often inferior or lateral); it is limited by the pericardium (3). Pseudoaneurysms may be large, contributing to heart failure, almost always contain a thrombus, and often rupture completely. They are repaired surgically.

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Thrombotic Complications

Recurrent Ischemia

Any chest pain that remains or recurs 12 to 24 hours after myocardial infarction may represent recurrent ischemia. Post-MI ischemic pain indicates that more myocardium is at risk of infarction. Usually, recurrent ischemia can be identified by reversible ST-T changes on the ECG; blood pressure may be elevated.

Recurrent ischemia is often silent (ECG changes without pain) (1), so serial ECGs are routinely performed every 8 hours for 1 day and then daily. Recurrent ischemia is treated similarly to unstable angina. Sublingual or IV nitroglycerin is usually effective. Coronary angiography and revascularization with percutaneous coronary intervention or coronary artery bypass grafting should be considered to salvage ischemic myocardium.

Mural Thrombosis

A study of patients with acute anterior MI treated with PCI found that 3.9% developed a left ventricular mural thrombosis during the hospitalization (2). Because risk is low, prophylaxis with anticoagulation is not always indicated (3).

Anticoagulant therapy may be considered for patients with STEMI and anterior wall akinesis or dyskinesis, but the patient's risk of bleeding must also be evaluated in light of dual antiplatelet therapy and resultant triple therapy should anticoagulation be chosen. Anticoagulants *are* recommended for patients after ACS with concomitant:

- Atrial fibrillation and high thromboembolic risk (eg, see table CHA2DS2-VASc score of ≥ 2)
- Mechanical heart valves
- Venous thromboembolism
- Hypercoagulable disorders

It also is considered reasonable to give anticoagulants to patients with STEMI and asymptomatic confirmed left ventricular mural thrombi, along with a P2Y12 inhibitor (4).

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Inflammatory Complications

Pericarditis

Pericarditis results from extension of myocardial necrosis through the wall to the epicardium; it is rare in the current era of reperfusion therapy (< 1%) (1).

A friction rub usually begins 24 to 96 hours after myocardial infarction onset. Earlier onset of the friction rub is unusual, although hemorrhagic pericarditis occasionally complicates the early phase of myocardial infarction. Acute tamponade is rare.

In addition to the friction rub, ECG shows diffuse ST-segment elevation and sometimes PR-interval depression. Echocardiography is frequently performed but is usually normal. Occasionally, small pericardial effusions and even unsuspected tamponade are detected.

Aspirin or another nonsteroidal anti-inflammatory drug (NSAID) usually relieves symptoms. Colchicine 0.5 to 1 mg orally once a day, alone, and especially added to conventional treatment, speeds recovery and helps prevent recurrences. High doses or prolonged use of NSAIDs or glucocorticoids may impair infarct healing and should be avoided; glucocorticoids may also increase the likelihood of recurrence (1). Anticoagulation is not contraindicated in early peri-infarction pericarditis but is contraindicated in later post-MI (Dressler) syndrome.

Post-MI Syndrome (Dressler Syndrome)

Post-MI syndrome develops rarely in patients several days to weeks or even months after acute myocardial infarction; incidence appears to have decreased in recent years (2, 3, 4). It is characterized by fever, pericarditis with a friction rub, pericardial effusion, pleuritis, pleural effusions, pulmonary infiltrates, and joint pain. This syndrome is caused by an autoimmune reaction to material from necrotic myocytes. It may recur.

Differentiating post-MI syndrome from extension or recurrence of infarction may be difficult. However, in post-MI syndrome, cardiac troponin does not increase significantly, and ECG changes are nonspecific.

NSAIDs (eg, aspirin 500 to 1000 mg orally every 6 to 8 hours until symptoms are relieved) are usually effective, but the syndrome can recur several times. Colchicine is effective for treatment and to prevent recurrences. In severe cases, a short, intensive course of another NSAID or a glucocorticoid may be necessary. High doses of an NSAID or a glucocorticoid are not used for more than a few days because they may interfere with early ventricular healing after an acute myocardial infarction.

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Drug Information for the Topic